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LLSMS2226 - Credit & Interest Rates Risk

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0. Introduction

« The key to risk management is never putting yourself in a position where you cannot live to fight another day. » (Attributed to Richard Fuld, former CEO and Chairman of Lehman Brothers). The irony of this statement became evident when Lehman Brothers failed to manage its own exposures, leading to its spectacular bankruptcy. This historic default served as a harsh wake-up call for the global financial system.

The 2008 financial crisis exposed important vulnerabilities in the way we price over-the-counter (OTC) derivatives. Before the collapse of Lehman Brothers, standard derivatives pricing assumed that major financial institutions were essentially risk-free entities. Valuations largely relied on a single-curve framework, assuming funding was always accessible at a universal, risk-free rate. However, the default of Lehman Brothers and the massive bailouts of other dominant market-makers, such as AIG, Citigroup, and Bank of America, demonstrated that counterparty credit risk is an unavoidable market reality, prompting a significant evolution in pricing rules. To account for these mutual default risks, the financial industry transitioned to a multi-curve framework and introduced mandatory Valuation Adjustments (xVA), specifically Credit and Debt Valuation Adjustments (CVA and DVA), to reflect the true, risk-adjusted value of OTC contracts.

In this project, we detail the quantitative framework developed by our xVA desk to compute the risk-adjusted price of a long equity forward contract. Our methodology relies on several core assumptions: a constant, continuously compounded risk-free rate of $r_f = 2\%$, a fixed recovery rate of 40% for both parties in the event of default, and an uncollateralized five-year OTC equity forward ($T = 5$, $K = 150$) as the underlying transaction. We develop our pricing engine in three logical steps. First, we establish our quantitative baseline by calibrating a Geometric Brownian Motion for the stock price and mean-reverting Vasicek processes for the default intensities. This process uses historical data to estimate correlations, while also bootstrapping current market CDS spreads to ensure our model remains consistent with current market prices. Next, we compute the analytical fair price and the initial Credit and Debt Valuation Adjustments (CVA and DVA) assuming these risk factors are independent. Finally, we move to a more realistic model by integrating our historical correlations via a Cholesky decomposition. This allows us to capture Wrong-Way Risk (WWR) through a joint Monte Carlo simulation. We then extend this framework by introducing a Credit Support Annex (CSA) to analyze how collateral mitigates these exposures, and we conclude with a sensitivity analysis of our final risk-adjusted valuation.

1. Data analysis and calibration

This section establishes the quantitative foundation for our pricing model. We begin by selecting and justifying the stochastic processes used to model the stock price and the default intensities of the bank and the counterparty. To utilize these models for actual derivative valuation, we must first estimate their governing parameters, such as volatility and drift, using historical market data. However, because empirical data is observed at discrete, fixed intervals, we cannot fit our continuous-time stochastic differential equations (SDEs) directly. To bridge this gap, we discretize these models into linear difference equations, allowing us to extract the precise annualized parameters via standard Ordinary Least Squares (OLS) regression. Once the individual parameters are calibrated, we compute the historical correlation matrix to capture the joint dependency structure between all risk factors. Finally, we utilize current market CDS data to construct deterministic shift functions, ensuring our simulated models perfectly replicate today's initial market curves and remain strictly arbitrage-free.

1.1. Choice and Justification of Stochastic Processes

In our framework, we select two standard stochastic processes for the stock price and the default risks that are both realistic and easy to implement. For the stock price S_t , we use a Geometric Brownian Motion (GBM), famously used by Black and Scholes ([Black & Scholes, 1973](#)). We chose this model because it ensures stock prices remain strictly positive, and it assumes that log-returns follow a normal distribution. This normality assumption is important for three practical reasons. First, the mathematics of normal distributions are completely solved, which allows us to use exact pricing formulas rather than estimates. Second, it makes time conversions simple; because normal variances add up perfectly, we can easily translate weekly volatility into yearly volatility. Finally, computers are highly optimized to generate normal random variables, ensuring our large-scale Monte Carlo simulations will run quickly and efficiently. However, this model has a well-known limitation: it assumes that volatility is constant over time, which does not reflect the volatility smile we observe in real markets. For the credit default intensities of both the Bank (B) and the Counterparty (C), we use the Vasicek model ([Vasicek, 1977](#)). This choice makes sense because credit spreads tend to be mean-reverting. If spreads jump during a crisis, they usually pull back to a long-term average over time. One drawback of the Vasicek model is that intensities can theoretically become negative because the model relies on a normal distribution. In our simulation, we simply fix this issue by flooring the intensities at zero, effectively truncating the intensity process and maintaining non-negative survival probabilities.

1.2. Mathematical Framework and OLS Calibration Methodology

1.2.1 Geometric Brownian Motion for the Stock Price

As we just mentioned, we model the underlying stock price using a Geometric Brownian Motion (GBM). Under the historical probability measure $\mathbb{P}$, the stock price thus follows this stochastic differential equation (SDE):

$$dS_t = \mu_S S_t dt + \sigma_S S_t dW_t^S \quad (1)$$

where μ_S is the annualized historical drift, σ_S is the annualized volatility, and dW_t^S is a standard Brownian motion increment which serves as the source of stochastic randomness.

Standard linear regression (OLS), which we use to extract our parameters, relies on the strict assumption of homoscedasticity, meaning the variance of the error terms must remain constant over time. The original SDE violates this assumption because its stochastic term ($\sigma_S S_t dW_t^S$) causes the absolute size of random price shocks to grow proportionally with the stock price S_t . If calibrated directly, OLS would be biased, overweighting periods when the stock price is high

simply because the absolute price swings are larger. To eliminate this scale dependency, we apply a logarithmic transformation by defining $f(S_t) = \ln(S_t)$. This isolates the constant volatility term ($\sigma_S dW_t^S$) required for OLS regression. Because the underlying stock price is driven by a random Brownian motion whose squared variation accumulates over time ($(dW_t^S)^2 = dt$), we must apply Itô's Lemma to accurately differentiate this log-price function and capture its full volatility profile (Shreve, 2004). The general expansion for the total differential $df(S_t)$ is given by:

$$d\ln(S_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial S_t} dS_t + \frac{1}{2} \frac{\partial^2 f}{\partial S_t^2} (dS_t)^2 \quad (2)$$

We calculate the partial derivatives of our function $f(S_t) = \ln(S_t)$ as follows:

$$\frac{\partial f}{\partial t} = 0, \quad \frac{\partial f}{\partial S_t} = \frac{1}{S_t}, \quad \frac{\partial^2 f}{\partial S_t^2} = -\frac{1}{S_t^2} \quad (3)$$

Substituting these derivatives into the general expansion formula yields:

$$d\ln(S_t) = \frac{1}{S_t} (\mu_S S_t dt + \sigma_S S_t dW_t^S) + \frac{1}{2} \left(-\frac{1}{S_t^2} \right) (\mu_S S_t dt + \sigma_S S_t dW_t^S)^2 \quad (4)$$

By applying standard Itô calculus multiplication rules, where $(dt)^2 = 0$, $dt \cdot dW_t^S = 0$, and $(dW_t^S)^2 = dt$, the squared term simplifies to $(dS_t)^2 = \sigma_S^2 S_t^2 dt$. Plugging this back into the equation reveals the continuous dynamics of the log-price:

$$d\ln(S_t) = \mu_S dt + \sigma_S dW_t^S - \frac{1}{2} \sigma_S^2 dt = \left(\mu_S - \frac{1}{2} \sigma_S^2 \right) dt + \sigma_S dW_t^S \quad (5)$$

Integrating this continuous-time relationship over a discrete weekly time step from $t-1$ to t gives the exact mathematical relation for the log-returns:

$$\int_{t-1}^t d\ln(S_u) = \int_{t-1}^t \left(\mu_S - \frac{1}{2} \sigma_S^2 \right) du + \int_{t-1}^t \sigma_S dW_u^S \quad (6)$$

$$\ln(S_t) - \ln(S_{t-1}) = \left(\mu_S - \frac{1}{2} \sigma_S^2 \right) dt + \sigma_S (W_t^S - W_{t-1}^S) \quad (7)$$

As shown in Equation 7, the weekly log-return $x_t = \ln(S_t) - \ln(S_{t-1})$ is cleanly split into a deterministic term and a stochastic term. The first term, $\left(\mu_S - \frac{1}{2} \sigma_S^2 \right) dt$, contains no randomness and therefore forms the exact mean (drift) of the distribution. The randomness is driven solely by the Brownian increment $(W_t^S - W_{t-1}^S)$. A fundamental property of Brownian motion is that increments over non-overlapping time intervals are strictly independent and follow a normal distribution $\mathcal{N}(0, dt)$. By the standard rules of variance, multiplying this noise by the constant volatility σ_S scales its variance to $\sigma_S^2 dt$. Consequently, our weekly historical log-returns x_t are mathematically guaranteed to be a sequence of independent and identically distributed (i.i.d.) normal random variables:

$$x_t \sim \mathcal{N} \left(\left(\mu_S - \frac{1}{2} \sigma_S^2 \right) dt, \sigma_S^2 dt \right) \quad (8)$$

Let $\hat{\mu}$ and $\hat{\sigma}$ be the sample mean and sample standard deviation computed directly from these historical log-returns series. Given our fixed weekly data frequency where $dt = \frac{1}{52}$, the continuous-time annualized parameters are calibrated using the following algebraic relationships:

$$\sigma_S = \frac{\hat{\sigma}}{\sqrt{dt}} \quad (9)$$

$$\mu_S = \frac{\hat{\mu}}{dt} + \frac{1}{2}\sigma_S^2 \quad (10)$$

The inclusion of the $\frac{1}{2}\sigma_S^2$ term is important as it serves as an Itô convexity adjustment. As established by Jensen's inequality, the expected value of the logarithm of a price series is strictly less than the logarithm of the expected price, a difference that financially represents volatility drag. This adjustment corrects for the curvature of the logarithmic function, allowing us to properly translate parameters from the sample space of log-returns back into the nominal expected returns (μ_S) required to value our equity forward contract (Hull, 2018).

1.2.2 Vasicek Process for Default Intensities

Before calibrating the credit risk parameters, the raw historical credit spreads s_t^X for $X \in \{B, C\}$ must be converted into proxy series of historical default intensities λ_t^X . Assuming a constant industry-standard recovery rate $R = 40\%$, the relationship is approximated by the standard credit triangle rule:

$$\lambda_t^X \approx \frac{s_t^X}{1 - R} \quad (11)$$

The default intensity is assumed to follow a continuous-time, mean-reverting Vasicek process:

$$d\lambda_t = \kappa(\theta - \lambda_t)dt + \eta dW_t^\lambda \quad (12)$$

where κ is the speed of mean reversion, θ is the long-term baseline mean, η is the instantaneous volatility of the intensity, and dW_t^λ is a standard Brownian motion error term.

Similar to our calibration approach for the Geometric Brownian Motion, estimating the Vasicek parameters using discrete weekly historical data requires an Ordinary Least Squares (OLS) framework. This once again necessitates establishing a bridge between the continuous-time SDE and a discrete-time statistical model. To establish this bridge, we compute the exact analytical solution of the continuous SDE over the interval $[t - 1, t]$. Because the drift term contains the state variable λ_t , we cannot integrate the equation directly to solve for it. To fix this problem, we multiply the entire equation by an integrating factor, $e^{\kappa t}$. This specific exponential function is chosen because it allows us to apply the reverse product rule, absorbing the problematic λ_t term into a single, clean derivative. By applying Itô's Lemma to the transformed function $f(t, \lambda_t) = \lambda_t e^{\kappa t}$, its stochastic differential yields:

$$\begin{aligned} d(\lambda_t e^{\kappa t}) &= \kappa \lambda_t e^{\kappa t} dt + e^{\kappa t} d\lambda_t \\ &= \kappa \lambda_t e^{\kappa t} dt + e^{\kappa t} [\kappa(\theta - \lambda_t)dt + \eta dW_t^\lambda] \\ &= \kappa \theta e^{\kappa t} dt + \eta e^{\kappa t} dW_t^\lambda \end{aligned} \quad (13)$$

As shown above, the terms involving λ_t on the right-hand side cancel out perfectly. This isolates the deterministic drift and the stochastic components, making the equation easy to integrate.

We now perform this integration over our discrete weekly time step, from $t - 1$ to t . To keep the notation clear, we use u as a dummy integration variable:

$$\int_{t-1}^t d(\lambda_u e^{\kappa u}) = \int_{t-1}^t \kappa \theta e^{\kappa u} du + \int_{t-1}^t \eta e^{\kappa u} dW_u^\lambda \quad (14)$$

We evaluate each component of this equation independently. Starting with the left-hand side, the Fundamental Theorem of Calculus dictates that the integral of the total differential represents the net structural variation of the process:

$$\int_{t-1}^t d(\lambda_u e^{\kappa u}) = \lambda_t e^{\kappa t} - \lambda_{t-1} e^{\kappa(t-1)} \quad (15)$$

Moving to the right-hand side, the first term is a standard Riemann integral of a deterministic exponential trend, which resolves directly to:

$$\int_{t-1}^t \kappa \theta e^{\kappa u} du = \theta [e^{\kappa u}]_{t-1}^t = \theta (e^{\kappa t} - e^{\kappa(t-1)}) \quad (16)$$

Finally, the second term on the right-hand side is an Itô stochastically driven integral. Because it depends on the random path of the Brownian motion, it cannot be simplified further and must remain in its functional integral form:

$$\int_{t-1}^t \eta e^{\kappa u} dW_u^\lambda \quad (17)$$

Substituting these components back into the integrated equation yields:

$$\lambda_t e^{\kappa t} - \lambda_{t-1} e^{\kappa(t-1)} = \theta (e^{\kappa t} - e^{\kappa(t-1)}) + \eta \int_{t-1}^t e^{\kappa u} dW_u^\lambda \quad (18)$$

To isolate the current intensity level λ_t , we multiply the entire equation by $e^{-\kappa t}$:

$$\lambda_t = \lambda_{t-1} e^{-\kappa(t-(t-1))} + \theta (1 - e^{-\kappa(t-(t-1))}) + \eta \int_{t-1}^t e^{-\kappa(t-u)} dW_u^\lambda \quad (19)$$

Since the time step corresponds to our constant discrete weekly interval ($t - (t-1) = dt = \frac{1}{52}$), the exact transition equation simplifies to:

$$\lambda_t = \lambda_{t-1} e^{-\kappa dt} + \theta (1 - e^{-\kappa dt}) + \eta \int_{t-1}^t e^{-\kappa(t-u)} dW_u^\lambda \quad (20)$$

We can directly map this continuous relationship onto a discrete-time Autoregressive model of order 1 (AR(1)), which allows us to estimate the process using standard linear regression:

$$\lambda_t = \alpha + \beta \lambda_{t-1} + \epsilon_t, \quad \text{where } \epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2) \quad (21)$$

In this mapped framework, the normally distributed error term ϵ_t corresponds exactly to the stochastic integral from our transition equation. To complete our statistical model, we must analytically determine the variance of this error term (σ_ϵ^2).

Let us define this stochastic component as $X = \eta \int_{t-1}^t e^{-\kappa(t-u)} dW_u^\lambda$. Since the expectation of an Itô integral with respect to a standard Brownian motion is strictly zero ($\mathbb{E}[X] = 0$), the variance simplifies directly to the expectation of its square. By applying Itô's Isometry ([Ibe, 2013](#)), the squared expectation of the stochastic integral reduces to a deterministic Riemann integral:

$$\begin{aligned} \text{Var}[X] &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \mathbb{E} \left[\left(\eta \int_{t-1}^t e^{-\kappa(t-u)} dW_u^\lambda \right)^2 \right] \\ &= \eta^2 \int_{t-1}^t (e^{-\kappa(t-u)})^2 du \\ &= \eta^2 \int_{t-1}^t e^{-2\kappa(t-u)} du \end{aligned}$$

To solve this deterministic integral, we apply the substitution $v = -2\kappa(t-u)$, which yields the exact closed-form expression for the discrete variance ([Glasserman, 2003](#)):

$$\sigma_\epsilon^2 = \frac{\eta^2}{2\kappa} (1 - e^{-2\kappa dt}) \quad (22)$$

Using standard OLS, we regress the weekly intensity series against its one-period lag to obtain the empirical coefficients α , β , and the residual standard error σ_ϵ . To extract the continuous annualized Vasicek parameters (κ, θ, η) , we simply match these empirical results back to the exact analytical solution derived earlier:

1. **Speed of mean reversion (κ):** By comparing the autoregressive terms, we observe that $\beta = e^{-\kappa dt}$. Taking the natural logarithm on both sides and isolating κ yields:

$$\kappa = -\frac{\ln(\beta)}{dt} \quad (23)$$

2. **Long-term baseline mean (θ):** At the long-term equilibrium, the expected intensity stabilizes such that $\mathbb{E}[\lambda_t] = \mathbb{E}[\lambda_{t-1}] = \theta$. Substituting θ into the deterministic part of the AR(1) equation gives $\theta = \alpha + \beta\theta$. Solving for θ results in:

$$\theta = \frac{\alpha}{1 - \beta} \quad (24)$$

3. **Volatility of the intensity (η):** The variance of the continuous stochastic integral over the interval dt is given by Equation 22. Setting this equal to the empirical residual variance from our regression (σ_ϵ^2) and solving for η provides the final conversion formula:

$$\eta = \sigma_\epsilon \sqrt{\frac{2\kappa}{1 - e^{-2\kappa dt}}} \quad (25)$$

where $dt = \frac{1}{52}$ represents our fixed weekly time increment expressed on an annual scale. This step ensures that our discrete statistical regression perfectly reflects the continuous Vasicek model.

The resulting annualized parameters extracted from our historical datasets for both the Bank (B) and the Counterparty (C) are summarized in Table 1.

Reference entity	Speed of reversion (κ)	Long-term mean (θ)	Volatility (η)
Bank (B)	3.2331	0.0062	0.0108
Counterparty (C)	1.4960	0.0143	0.0077

Table 1: Calibrated continuous annualized Vasicek parameters via OLS

These three extracted parameters (κ , θ , and η) are now ready to be injected into our continuous-time framework. They will drive the deterministic shift adjustments $\phi(t)$ and feed the joint Monte Carlo simulation paths.

1.3. Analysis of Calibration Outputs and Plots

Before using our calibrated parameters to price the equity forward contract and assess the credit risk, it is essential to visually inspect our historical data and validate the model fits. We evaluate both our equity and default intensities models by benchmarking their historical trajectories against 90% confidence intervals, generated via Monte Carlo simulations of 1,000 random paths.

1.3.1 Equity Model: Geometric Brownian Motion (GBM)

Figure 1 illustrates the historical stock price benchmarked against its simulated GBM confidence tunnel. The empirical trajectory closely tracks the theoretical median (dashed blue line) and remains well-contained within the confidence bounds throughout the five-year period. While the actual price briefly breaches the upper boundary during a strong market rally between late 2021 and mid-2022, it quickly reverts and ranges close to the same level as the median until the end of the timeline. This confirms that our calibrated parameters successfully capture the asset's long-term structural dynamics and volatility, validating the use of the GBM framework.

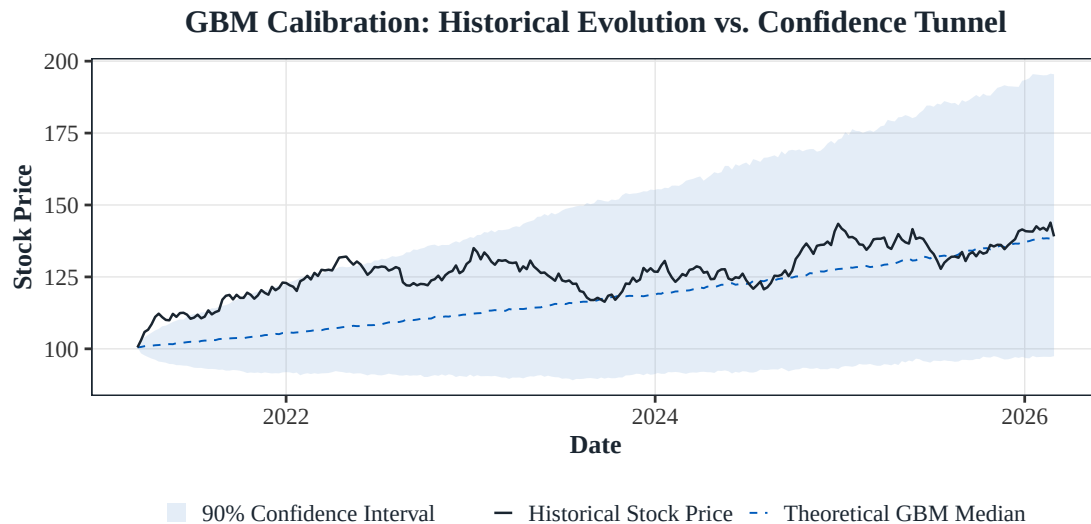
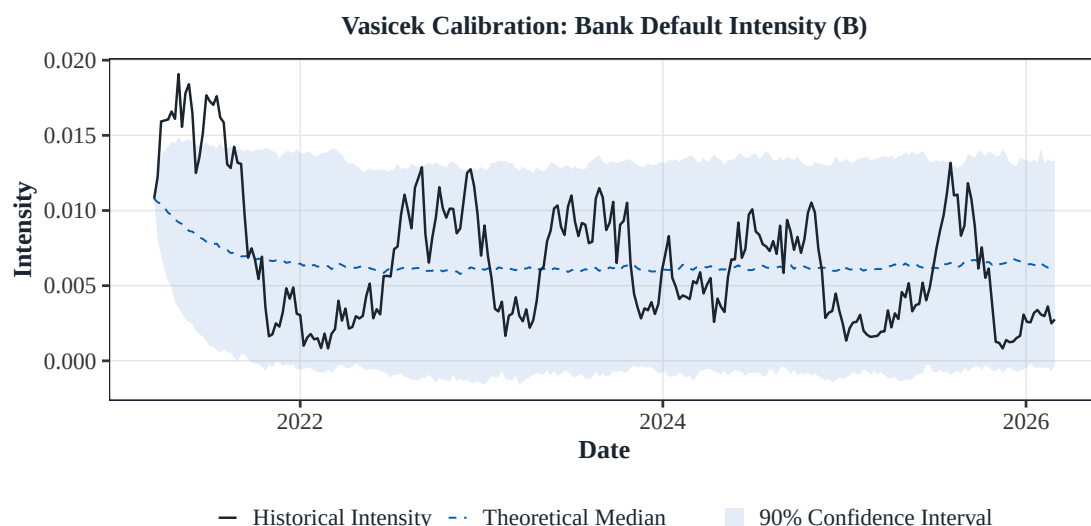


Figure 1: GBM validation model showing the historical stock price (black curve) benchmarked against the simulated asset median trajectory (blue dashed line) and the 90% confidence interval region (shaded light blue zone).

1.3.2 Credit Model: Vasicek Intensity Process

Figure 2 applies this same validation framework to our Vasicek calibration. In contrast to the expanding GBM tunnel, the Vasicek confidence intervals flatten into stationary bounds, reflecting the mean-reverting nature of this process.

The historical default intensities for both the Bank and the Counterparty are, except for the beginning, consistently captured within these boundaries, confirming that our parameters (κ, θ, η) accurately model the stationary dynamics and structural limits of these credit risks.



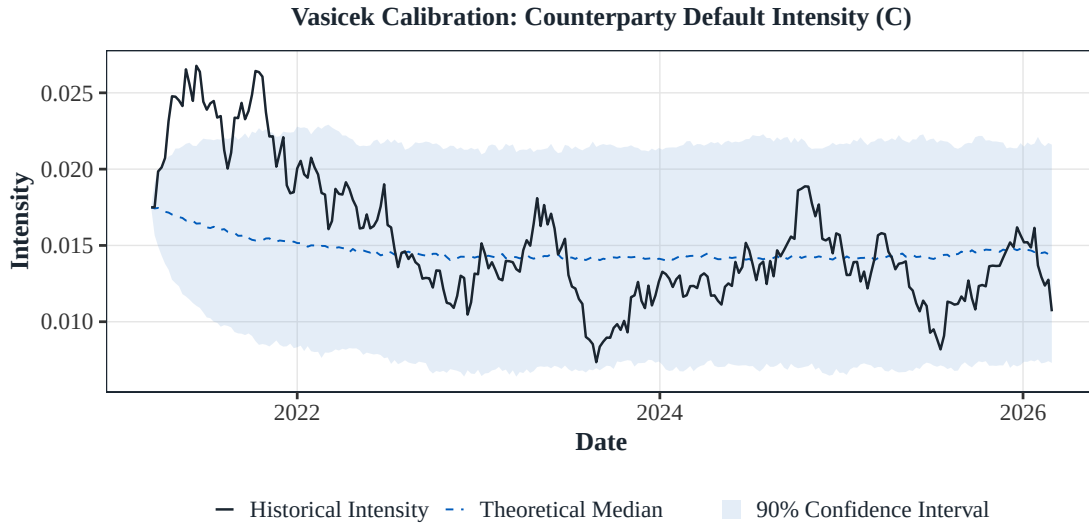


Figure 2: Vasicek validation: historical intensities for the Bank (top) and Counterparty (bottom) vs. simulated medians and bounded 90% confidence intervals.

1.3.3 Residual Diagnostics

Confidence tunnels show the general shape of the simulated paths, but we must also check the model residuals. We extracted the shocks from the Geometric Brownian Motion (ϵ^S) and the residuals from the two Vasicek processes (ϵ^B, ϵ^C). As shown in the Normal Q-Q plots (Figure 3), the middle data points follow the straight normal line very well. There are small deviations in the extreme tails, which is very common in financial data, but the normal assumption remains solid for most of the data. Additionally, the residuals show no major autocorrelation. This proves our models successfully captured the historical time trends. Overall, these checks confirm our models are statistically sound before we build the correlation matrix.

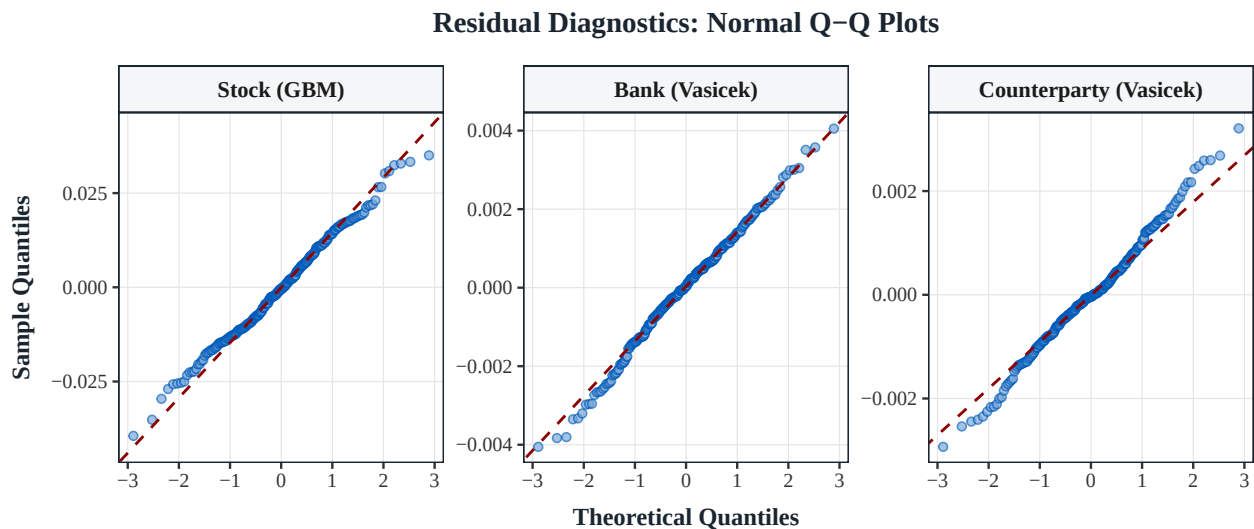


Figure 3: Normal Q-Q plots for the stock returns and the default intensity residuals.

1.4. Estimation of the Historical Correlation Matrix

In global OTC derivatives markets, equity prices and credit spreads are heavily interconnected. To capture these co-movements and accurately model the joint behavior of our risk factors, we estimate a historical correlation matrix based on the random shocks extracted from our previous calibrations.

Before analyzing their interconnection, it is highly instructive to compare the baseline risk profiles of our two credit entities. Figure 4 displays the historical evolution of the weekly default intensities over the five-year sample window.

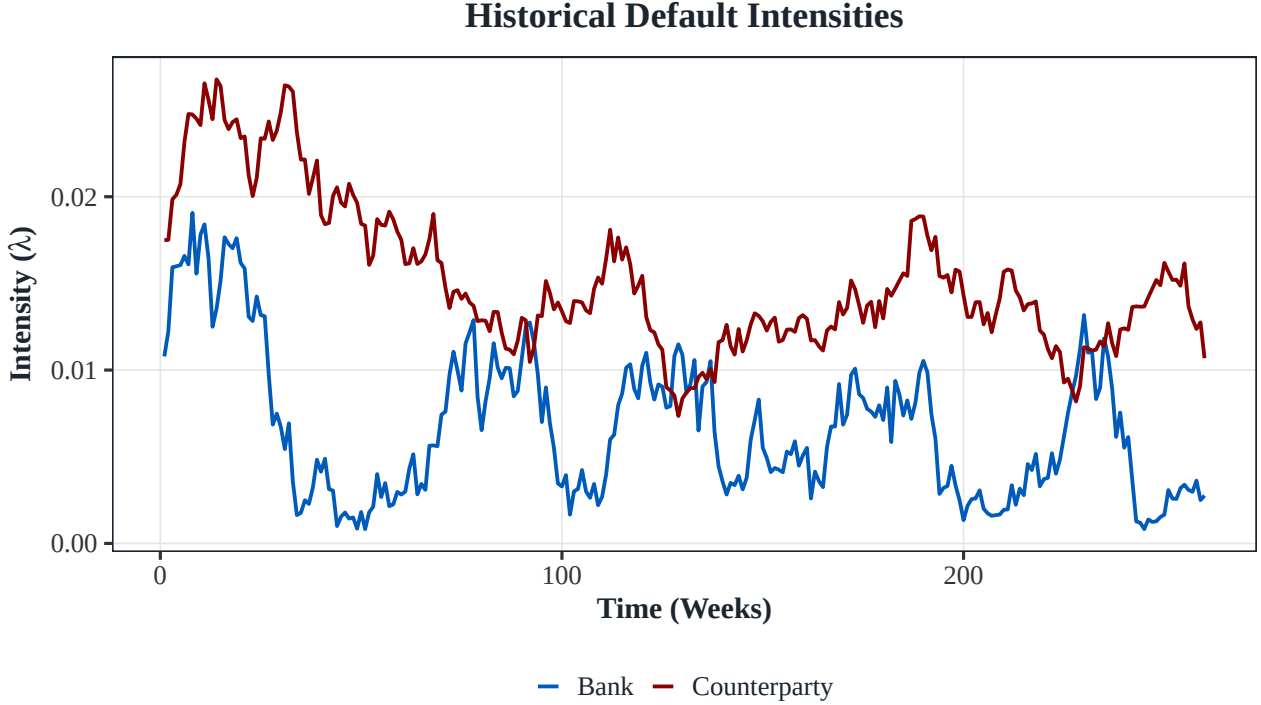


Figure 4: Historical evolution of the weekly default intensities for the Bank (blue) and the Counterparty (red) over the 5-year data window.

The Counterparty's default intensity is structurally higher than the Bank's across the entire timeline, fluctuating around a notably higher baseline equilibrium. This visual gap indicates that the market penalizes the Counterparty with consistently wider credit spreads, marking it as a significantly riskier entity. To understand how this risk interacts with the equity market, we must now strip away these baseline trends to analyze the pure correlation of their shocks.

1.4.1 Mathematical Extraction of Statistical Shocks

To construct our correlation matrix, we must isolate the pure random components of each process by stripping away their deterministic trends (i.e., the drift for the stock and the mean-reverting pull for credit intensities). Under our continuous-time setup, these uncertainties originate from the standard Brownian motions (W_t^S , W_t^B , and W_t^C). By evaluating our processes over a discrete weekly step $dt = \frac{1}{52}$, we isolate the true underlying historical shocks (ϵ_t) by computing the residuals of our calibrated models. This ensures our correlation analysis captures only the unexpected, co-moving market innovations:

1. **Equity shocks (ϵ_t^S):** Recalling the discrete-time expression obtained via Itô's lemma in Equation (7), the random shock corresponds exactly to the scaled Brownian increment. We isolate it by subtracting the expected theoretical drift from the observed log-return:

$$\epsilon_t^S = \sigma_S \left(W_t^S - W_{t-1}^S \right) = \underbrace{[\ln(S_t) - \ln(S_{t-1})]}_{\text{Observed Log-Return}} - \underbrace{\left(\mu_S - \frac{1}{2}\sigma_S^2 \right) dt}_{\text{Expected Theoretical Drift}} \quad (26)$$

These mean-centered residuals capture the weekly unexpected equity market movements.

2. **Credit intensity shocks (ϵ_t^B and ϵ_t^C):** For the mean-reverting Vasicek intensities, the discrete noise term relates theoretically to the continuous stochastic integral. Empirically, these shocks are isolated as the error terms from our discrete AR(1) OLS regressions. By subtracting the predicted next-period intensity from the observed value, we extract the unexplained portion of the spread:

$$\epsilon_t^B = \lambda_t^B - (\alpha^B + \beta^B \lambda_{t-1}^B) \quad (27)$$

$$\epsilon_t^C = \lambda_t^C - (\alpha^C + \beta^C \lambda_{t-1}^C) \quad (28)$$

These residuals capture sudden, unexpected shifts in credit risk that cannot be explained by historical mean reversion alone.

1.4.2 The Pearson correlation framework

Given our aligned column vectors of shocks for a sample size of $N - 1$ weekly steps, we measure the linear dependence between any two shock series ϵ^i and ϵ^j (where $i, j \in \{S, B, C\}$) using the sample Pearson correlation coefficient $\rho_{i,j}$:

$$\rho_{i,j} = \frac{\text{Cov}(\epsilon^i, \epsilon^j)}{\sqrt{\text{Var}(\epsilon^i)\text{Var}(\epsilon^j)}} = \frac{\sum_{t=2}^N (\epsilon_t^i - \bar{\epsilon}^i)(\epsilon_t^j - \bar{\epsilon}^j)}{\sqrt{\sum_{t=2}^N (\epsilon_t^i - \bar{\epsilon}^i)^2 \sum_{t=2}^N (\epsilon_t^j - \bar{\epsilon}^j)^2}} \quad (29)$$

Because our OLS residuals and equity returns are mean-centered, their empirical mean is approximately zero ($\bar{\epsilon} \approx 0$). This simplifies the formula directly to the normalized inner product of the shock vectors. Applying this to our empirical data yields the following symmetric 3×3 historical correlation matrix ρ :

$$\rho = \begin{matrix} & \begin{matrix} \text{Stock} \\ \text{Bank intensity} \\ \text{Counterparty intensity} \end{matrix} \end{matrix} \begin{pmatrix} \text{Stock} & \begin{pmatrix} 1.0000 & -0.0108 & 0.5155 \\ -0.0108 & 1.0000 & 0.2257 \\ 0.5155 & 0.2257 & 1.0000 \end{pmatrix} \end{pmatrix} \quad (30)$$

where the diagonal elements are naturally equal to 1, representing the perfect linear correlation of each risk process with itself.

This matrix highlights three key market dynamics. First, the correlation between the stock price and the Bank's credit shocks is practically zero ($\rho_{S,B} = -0.0108$). This suggests that the Bank's default intensity behaves independently of the equity asset's performance. Second, there is a weak positive relationship between the two credit intensities ($\rho_{B,C} = 0.2257$), indicating a light co-movement likely driven by broader macroeconomic factors affecting the credit market. Finally, and most importantly, the matrix establishes a strong positive correlation ($\rho_{S,C} = 0.5155$) between the stock price and the Counterparty's credit shocks. This specific configuration mathematically validates the presence of Wrong-Way Risk (WWR) within our framework. As defined by Gregory (2015), WWR represents an unfavourable dependence where exposure is highest precisely when the counterparty is most likely to default. In our context, because the Bank is long the forward contract, an unexpected upward jump in the stock price increases our replacement cost (exposure). Due to the positive correlation, this increase in exposure tends to coincide simultaneously with a sudden spike in the Counterparty's credit spread (deteriorating credit quality), exposing the Bank to significant counterparty credit risk.

To visually corroborate this critical dynamic, Figure 5 maps the underlying equity return shocks against the Counterparty's credit shocks.

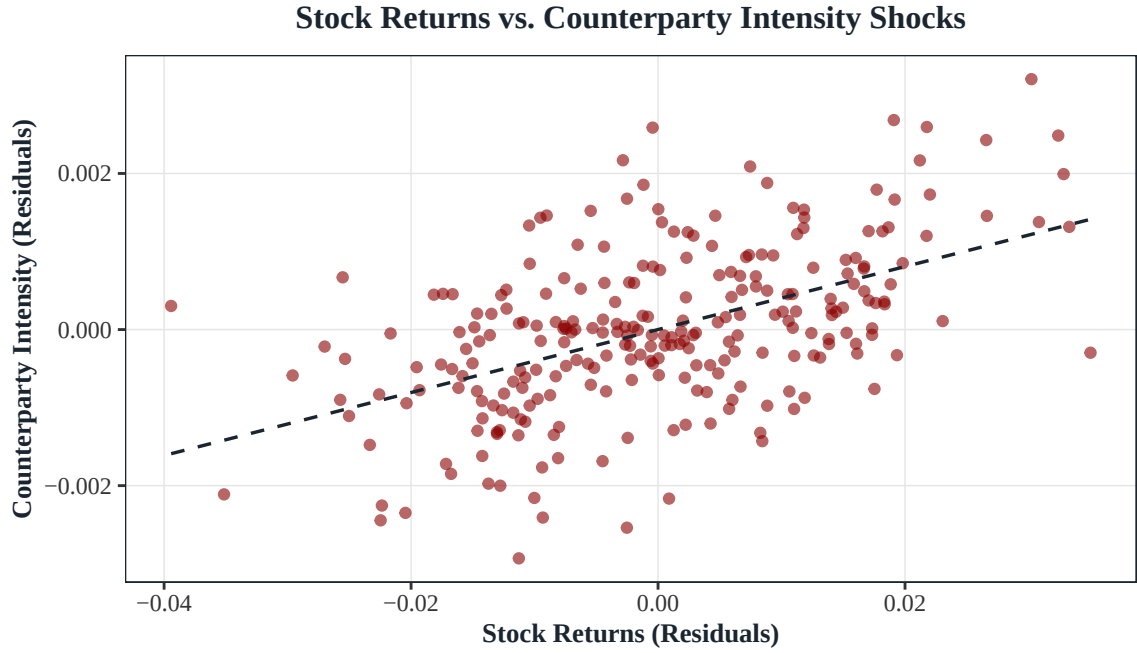


Figure 5: Statistical scatter plot mapping the underlying stock return residuals (ϵ_t^S) against the Counterparty's default intensity residuals (ϵ_t^C), fitted with an OLS linear trendline.

The ordinary least squares regression line fitted through the data points exhibits a clear upward trajectory. This plot clearly supports the historical correlation coefficient, proving that our WWR framework relies on real-world data.

1.5. Credit curve bootstrapping under piecewise constant intensity

To accurately evaluate counterparty credit risk and compute credit valuation adjustments, we must align our continuous pricing models with how the market currently prices the risk of default. This is achieved by extracting the market-implied survival probability curves, $G^{B, mkt}(0, t)$ and $G^{C, mkt}(0, t)$, from the Credit Default Swap (CDS) term structures provided in Table 2.

1.5.1 Theoretical foundations and leg formulations

A Credit Default Swap (CDS) can be seen as an insurance contract against the default of a reference entity over a defined time horizon T , protecting a specified notional amount N . The market does not directly quote the probability of default of an entity, but it provides us the price to protect us from the default. Because CDS markets are highly liquid and isolate pure credit risk from interest rate risk, they are the industry standard for extracting market-implied, risk-neutral survival probabilities.

Mechanically, the contract comprises two opposing cash flow streams:

- **The Premium Leg (PL):** Regular, periodic spread payments made by the protection buyer to maintain the insurance.
- **The Default Leg (DL):** A contingent payout made by the protection seller to cover the Loss Given Default (LGD) if, and only if, a credit event occurs.

By fundamental market convention, the theoretical par spread $s(T_m)$ for a newly initialized contract with maturity pillar T_m is established such that the Net Present Value (NPV) of the contract at inception ($t = 0$) is exactly zero. This equilibrium ensures that the present value of the expected premium payments perfectly matches the present value of the expected default

losses. Structuring the contract this way is highly efficient, as it eliminates the need for an upfront cash exchange between the counterparties to initiate the trade.

Consequently, to extract our survival probabilities, we rely on this strictly balanced, zero-NPV pricing condition:

$$\text{NPV} = \text{DL}(0, T_m) - \text{PL}(0, T_m) = 0 \implies \text{DL}(0, T_m) = \text{PL}(0, T_m) \quad (31)$$

Assuming continuous premium accumulation and a constant, continuously compounded risk-free rate r , the present value of the Premium Leg (PL) integrates these discounted payments over time, weighted by the probability $G(0, t)$ that the entity is still alive to make the payment:

$$\text{PL}(0, T_m) = s(T_m) \int_0^{T_m} e^{-rt} G(0, t) dt \quad (32)$$

Conversely, the present value of the Default Leg (DL) represents the expected discounted payout. It multiplies the Loss Given Default ($\text{LGD} = 1 - R$, where $R = 40\%$ is the assumed constant recovery rate) by the infinitesimal probability of the entity surviving to time t and then defaulting exactly in the next instant dt :

$$\text{DL}(0, T_m) = (1 - R) \int_0^{T_m} e^{-rt} (-dG(0, t)) = (1 - R) \int_0^{T_m} e^{-rt} \lambda(t) G(0, t) dt \quad (33)$$

To solve these integrals from discrete market quotes, we apply the standard financial assumption that the instantaneous default intensity $\lambda(t)$ is a piecewise constant function over each maturity bracket $t \in (T_{i-1}, T_i]$:

$$\lambda(t) = \lambda_i, \quad \forall t \in (T_{i-1}, T_i] \quad (34)$$

Consequently, the market survival probability function $G(0, t)$ can be formulated cumulatively for any time t falling within the m -th interval ($T_{m-1} < t \leq T_m$) as:

$$G(0, t) = \exp\left(-\int_0^t \lambda(u) du\right) = \exp\left(-\underbrace{\sum_{i=1}^{m-1} \lambda_i (T_i - T_{i-1})}_{\text{Completed full intervals}} - \underbrace{\lambda_m (t - T_{m-1})}_{\text{Current partial interval}}\right) \quad (35)$$

1.5.2 Step-by-step analytical derivation of the bootstrap algorithm

The ultimate objective of the bootstrap algorithm is to extract the unknown market default intensities, λ_m , for a newly observed CDS maturity pillar T_m . To find this value, we must enforce the zero-NPV condition, which dictates that the present value of the Premium Leg must exactly equal the present value of the Default Leg ($\text{PL} = \text{DL}$).

Recall that the formulas established previously for both legs involve integrating over the entire lifespan of the contract, from $t = 0$ up to T_m . However, because we established that the default intensity $\lambda(t)$ is piecewise constant, its value changes abruptly at the boundary of every market pillar. Consequently, we cannot compute the integral in one single continuous sweep. Instead, using the additive property of integrals, we must split the full integral $\int_0^{T_m}$ into a sequence of smaller, adjacent intervals that match the market brackets: $\int_0^{T_1} + \int_{T_1}^{T_2} + \dots + \int_{T_{m-1}}^{T_m}$.

If we closely examine the equations for both the Premium and Default legs, we see that their core mathematical engine relies on evaluating the exact same term: the discounted survival probability integral, $\int e^{-rt} G(0, t) dt$.

Therefore, rather than attempting to solve the whole equation at once, our first mathematical step is to isolate and solve this shared integral for just *one* generic interval segment $[T_{i-1}, T_i]$. We develop this core integral step-by-step to demonstrate its closed-form solution. First, we expand the survival probability $G(0, t)$ inside the interval using our cumulative formulation from Equation (35):

$$\int_{T_{i-1}}^{T_i} e^{-rt} G(0, t) dt = \int_{T_{i-1}}^{T_i} e^{-rt} \left[G(0, T_{i-1}) e^{-\lambda_i(t-T_{i-1})} \right] dt \quad (36)$$

Next, we pull all the constant terms (everything that does not depend on time t) outside the integral:

$$= G(0, T_{i-1}) e^{\lambda_i T_{i-1}} \int_{T_{i-1}}^{T_i} e^{-(r+\lambda_i)t} dt \quad (37)$$

We then compute the standard mathematical integral of the exponential function:

$$= G(0, T_{i-1}) e^{\lambda_i T_{i-1}} \left[\frac{e^{-(r+\lambda_i)t}}{-(r+\lambda_i)} \right]_{T_{i-1}}^{T_i} \quad (38)$$

Evaluating this at the upper and lower bounds (T_i and T_{i-1}), and using the negative sign to flip the subtraction order, gives:

$$= G(0, T_{i-1}) e^{\lambda_i T_{i-1}} \left(\frac{e^{-(r+\lambda_i)T_{i-1}} - e^{-(r+\lambda_i)T_i}}{r + \lambda_i} \right) \quad (39)$$

To simplify, we factor out $e^{-(r+\lambda_i)T_{i-1}}$ from the numerator to isolate the time difference inside the bracket:

$$= G(0, T_{i-1}) e^{\lambda_i T_{i-1}} e^{-(r+\lambda_i)T_{i-1}} \left(\frac{1 - e^{-(r+\lambda_i)(T_i-T_{i-1})}}{r + \lambda_i} \right) \quad (40)$$

Finally, we cancel out the opposing $\lambda_i T_{i-1}$ terms and substitute $\Delta T_i = T_i - T_{i-1}$ to represent the interval length:

$$= e^{-rT_{i-1}} G(0, T_{i-1}) \left[\frac{1 - e^{-(r+\lambda_i)\Delta T_i}}{r + \lambda_i} \right] \quad (41)$$

Having solved the math for a single segment, we can now express the full Premium Leg and Default Leg as the sum of these individual segments from $i = 1$ up to m . By substituting this solved segment back into our pricing models and setting $PL = DL$, the zero-NPV condition becomes a purely algebraic equation:

$$\underbrace{s(T_m) \sum_{i=1}^m e^{-rT_{i-1}} G(0, T_{i-1}) \left[\frac{1 - e^{-(r+\lambda_i)\Delta T_i}}{r + \lambda_i} \right]}_{\text{Total Premium Leg (PL)}} = \underbrace{(1 - R) \sum_{i=1}^m \lambda_i e^{-rT_{i-1}} G(0, T_{i-1}) \left[\frac{1 - e^{-(r+\lambda_i)\Delta T_i}}{r + \lambda_i} \right]}_{\text{Total Default Leg (DL)}} \quad (42)$$

This equation forms the basis of our bootstrapping process. Because we solve the maturities step-by-step, all the earlier default intensities ($\lambda_1, \lambda_2, \dots, \lambda_{m-1}$) are already known numbers. As a result, λ_m is the only missing piece left in the entire equation. We simply solve it to find the exact value of λ_m that makes both sides equal. Once we find it, we save this value and move forward to the next time period.

1.5.3 Analysis of bootstrapped curves and market inputs

Before calculating the continuous survival curves, we first take a look at the raw market inputs. Table 2 summarizes the current Credit Default Swap (CDS) spreads for both the Bank (B) and the Counterparty (C) across different maturities.

Maturity T (Years)	Bank spread (bps)	Counterparty spread (bps)
1	15	55
3	20	72
5	23	80
7	26	90
10	30	105

Table 2: Current market CDS spread term structures (in basis points)

This data reveals that the Counterparty exhibits a significantly higher credit risk profile than the Bank. At every single time horizon, the market demands a much higher premium to insure against the Counterparty's default. We can also see that the cost of protection increases over time for both entities, but it accelerates much faster for the Counterparty. For instance, the difference in risk between the two parties is only 40 basis points in the short term (1 year). However, as time goes on, this credit gap steadily widens, reaching a 75 basis point difference by year 10. This clearly demonstrates that the market associates long-term exposure to the Counterparty with a high degree of uncertainty.

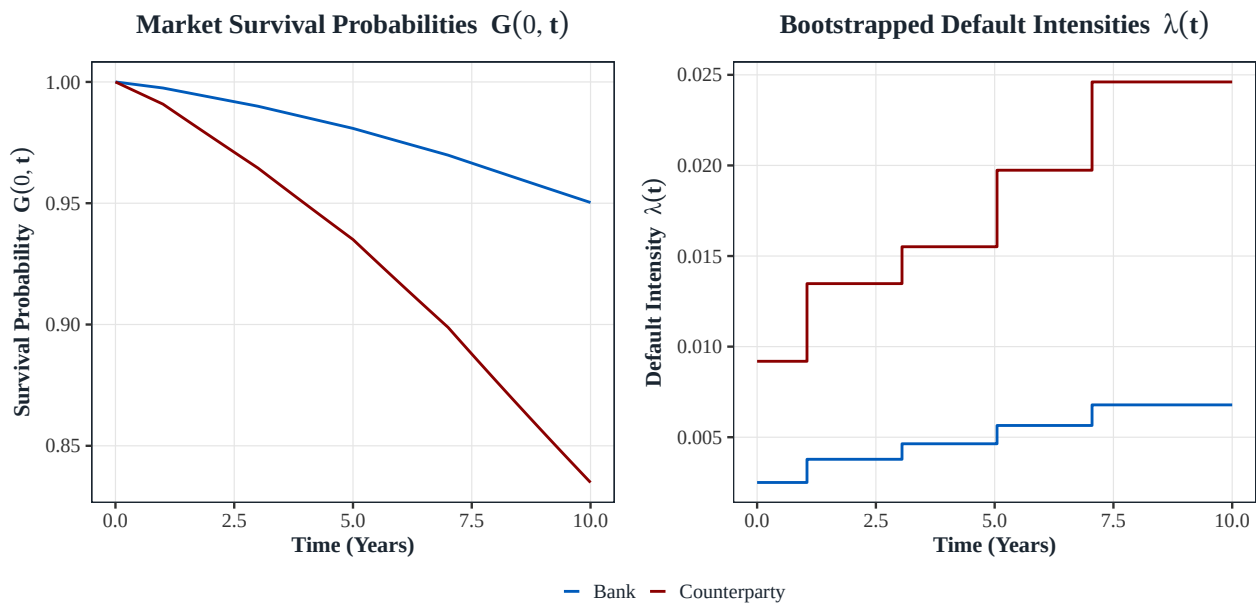


Figure 6: Bootstrapped market credit curves: Survival probabilities $G^{mkt}(0, t)$ (left panel) and corresponding piecewise constant default intensities $\lambda(t)$ (right panel) for the Bank and the Counterparty.

Once the bootstrapping algorithm is complete, we can visualize the final results in Figure 6. The left panel displays the market survival probabilities, $G^{mkt}(0, t)$. As expected, both curves drop steadily over time because the longer a contract lasts, the higher the cumulative chance that a default will eventually occur. Matching our analysis of the CDS spreads, the Counterparty's survival curve falls much faster than the Bank's. By the 10-year mark, the market prices the

Counterparty's chance of survival below 85%, whereas the Bank remains highly secure above 95%. The right panel illustrates the extracted default intensities, $\lambda(t)$. This chart perfectly highlights the staircase effect created by our piecewise constant assumption. At each market pillar (1, 3, 5, 7, and 10 years), the risk level jumps and then stays flat until the next pillar. For both entities, we observe a clear upward-sloping trend. This is very standard as the financial and macroeconomic uncertainty surrounding any company is much higher a decade from now than it is for the upcoming year.

1.6. Determination of the deterministic shift functions

At this stage in the calibration process, we face a fundamental inconsistency. On one hand, our calibrated Vasicek processes (y_t^B, y_t^C) successfully capture the dynamic behavior of past spread movements, specifically their volatility (η) and mean reversion speed (κ). On the other hand, because these parameters are purely historical, a standard Vasicek model will fail to perfectly re-price today's actual market term structure that we derived in the previous section. If our theoretical pricing model disagrees with the current CDS market quotes, it immediately introduces pricing arbitrage.

To resolve this conflict, we must bridge the gap between historical dynamics and current market reality. We achieve this by extending our setup into a Hull-White framework (Hull & White, 1990). This approach introduces a deterministic shift function, $\phi(t)$, a time-dependent parameter used to calibrate the model to the current market credit curve. It shifts the mean of the Vasicek process up or down at every future time step so that the model's initial survival probabilities perfectly match the bootstrapped market curves. Therefore, the total market-adjusted default intensity is modeled as the sum of this deterministic market correction and the stochastic historical process:

$$\lambda_t^X = \phi^X(t) + y_t^X, \quad X \in \{B, C\} \quad (43)$$

1.6.1 Mathematical derivation of the exact closed-form shift function

Our model now incorporates a random Vasicek process. Consequently, we must apply standard stochastic credit pricing theory. In this framework, the survival probability is defined as the expected value of the exponentially discounted cumulative default intensity. Therefore, the core requirement of our Hull-White calibration is to ensure that this model-derived expectation is identically equal to the bootstrapped market survival curve for any future maturity t :

$$G_0^{model}(t) = G^{mkt}(0, t) \implies \mathbb{E} \left[\exp \left(- \int_0^t (\phi(u) + y_u) du \right) \right] = G^{mkt}(0, t) \quad (44)$$

Because the expectation operator $\mathbb{E}[\cdot]$ only affects stochastic variables, and $\phi(t)$ is a purely deterministic function, its integral can be taken out of the expectation:

$$\exp \left(- \int_0^t \phi(u) du \right) \cdot \mathbb{E} \left[\exp \left(- \int_0^t y_u du \right) \right] = G^{mkt}(0, t) \quad (45)$$

Let $G_0^y(t) = \mathbb{E} \left[\exp \left(- \int_0^t y_u du \right) \right]$ represent the base survival probability generated solely by the standard Vasicek process. We intentionally start this base process at an initial value of $y_0 = 0$, as all of today's initial market risk is fully absorbed by the deterministic shift. To algebraically isolate our unknown function $\phi(t)$, we take the natural logarithm on both sides, which transforms the complex exponentials into a simple linear relationship:

$$- \int_0^t \phi(u) du + \ln(G_0^y(t)) = \ln(G^{mkt}(0, t)) \quad (46)$$

Next, we differentiate this continuous equation with respect to the maturity parameter t . By applying the Fundamental Theorem of Calculus, the derivative of the integrated shift function exactly yields $-\phi(t)$:

$$-\phi(t) + \frac{\partial \ln(G_0^y(t))}{\partial t} = \frac{\partial \ln(G^{mkt}(0, t))}{\partial t} \quad (47)$$

By financial definition, the negative derivative of a log-survival curve represents the instantaneous forward default intensity, denoted as $f(0, t) = -\frac{\partial \ln G(0, t)}{\partial t}$. Substituting these standard forward rates transforms our differential equation into a simple subtraction:

$$-\phi(t) - f^y(0, t) = -f^{mkt}(0, t) \implies \phi(t) = f^{mkt}(0, t) - f^y(0, t) \quad (48)$$

To compute the final formula, we must evaluate both of these forward rates. For a standard Vasicek process starting at zero, the mathematics of affine term structure models provides a well-known, exact analytical solution for the continuous forward rate of the base process (Brigo & Mercurio, 2006):

$$f^y(0, t) = \theta \left(1 - e^{-\kappa t}\right) - \frac{\eta^2}{2\kappa^2} \left(1 - e^{-\kappa t}\right)^2 \quad (49)$$

Regarding the market survival curve, because our bootstrapped default intensities are piecewise constant, the market's instantaneous forward rate $f^{mkt}(0, t)$ simplifies directly to the discrete step value λ_m corresponding to the maturity interval containing t . Combining these two components results in the final, closed-form formula for the deterministic shift function:

$$\phi(t) = f^{mkt}(0, t) - \left[\theta \left(1 - e^{-\kappa t}\right) - \frac{\eta^2}{2\kappa^2} \left(1 - e^{-\kappa t}\right)^2 \right] \quad (50)$$

1.6.2 Graphical analysis of the shift functions

Now that we have derived the exact closed-form formula, we can compute the deterministic shift trajectories over a standard 10-year horizon. These functions act as a time-dependent adjustment. They shift the standard Vasicek model so that it perfectly matches the actual credit curves we see in the market today. The resulting calibrated shifts for both the Bank and the Counterparty are illustrated in Figure 7.

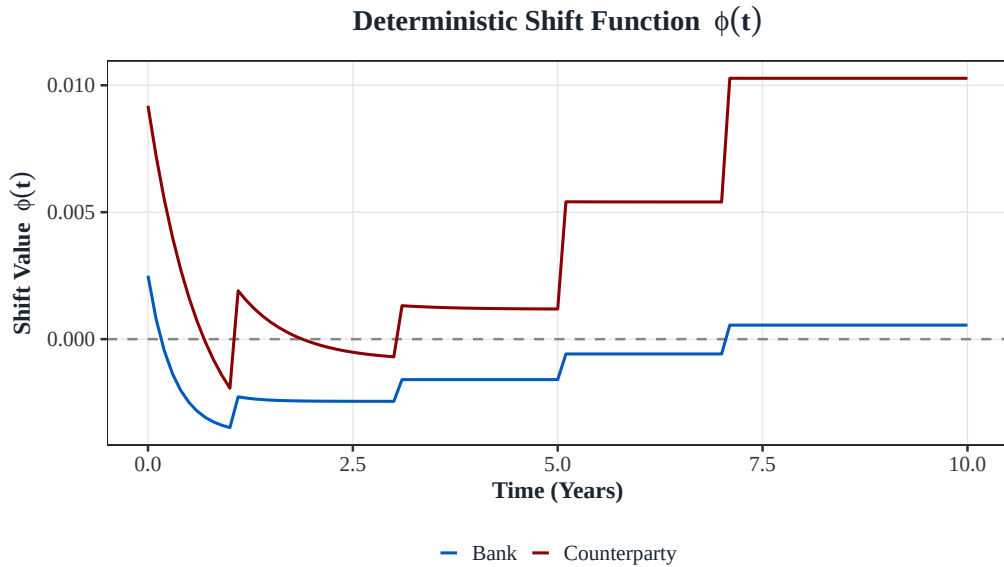


Figure 7: Calibrated deterministic shift functions $\phi(t)$ over a 10-year horizon for the Bank (blue curve) and the Counterparty (red curve).

Looking at these curves reveals two key characteristics. First, because the Counterparty’s bootstrapped market credit curve is steeper and significantly higher than the Bank’s, its corresponding shift function $\phi^C(t)$ (red curve) is positioned much higher on the vertical axis. This demonstrates that a much larger mathematical adjustment is required to bridge the gap between the Counterparty’s historical baseline and its current market risk premium. Second, both trajectories exhibit sharp, visible jumps. These discontinuities directly reflect the transitions between the discrete maturities (1, 3, 5, 7, and 10 years) of the underlying CDS quotes from Table 2. By incorporating these sudden jumps, the shift functions successfully and continuously correct for any discrepancies between past historical behavior and today’s actual credit risk pricing.

1.7. Conclusion

This section established the quantitative foundation of our pricing model by linking historical data with forward-looking market expectations. We first extracted the baseline behavior of our core risk factors by fitting a Geometric Brownian Motion (GBM) to stock returns and mean-reverting Vasicek processes to historical default intensities. This historical analysis was crucial to capture how these variables naturally interact over time. By analyzing the statistical residuals, we uncovered a clear positive correlation ($\rho_{S,C} > 0$) between equity returns and the Counterparty’s credit shocks. Because the Bank holds a long position in the equity forward contract, this dynamic mathematically confirms the presence of Wrong-Way Risk (WWR). It proves that our financial exposure will increase exactly when the Counterparty’s credit quality is deteriorating.

However, relying solely on historical data is insufficient for derivative pricing because it ignores how the market currently values risk. To correct this, we bootstrapped CDS spreads to extract today’s market-implied default intensities. We then reconciled our historical models with these current prices by utilizing a Hull-White extension. By deriving the exact deterministic shift function, $\phi(t) = f^{mkt}(0, t) - f^y(0, t)$, we forced our theoretical models to perfectly respect today’s market reality, strictly preventing any pricing arbitrage.

Now that the stochastic parameters are calibrated, the correlation matrix is estimated, and the deterministic shifts are calculated, our theoretical framework is complete. In the following sections, we will combine these aligned models to build a joint Monte Carlo simulation. This simulation will allow us to first price the baseline equity forward contract, and subsequently compute the value adjustments (xVA).

2. Fair price and valuation adjustments under independence

With the stochastic framework fully calibrated, we now apply these models to the valuation of the OTC equity forward contract. The primary objective of this section is to compute the base price of the derivative and quantify its bilateral credit risk through the Credit Valuation Adjustment (CVA) and Debt Valuation Adjustment (DVA).

Crucially, we perform these initial calculations under the strict assumption of statistical independence, meaning we temporarily force the correlation between the stock price and the default intensities to zero ($\rho = 0$). Establishing this uncorrelated baseline provides a clean reference point that will allow us to precisely isolate and measure the specific financial impact of Wrong-Way Risk (WWR) when the historical correlation is reintroduced later in the thesis.

To achieve this, we first define the contract’s payoff and exposure profiles. We then formalize the mathematical xVA pricing equations and use an independent Monte Carlo simulation to compute the baseline valuation metrics.

2.1. Analytical risk-free fair price of the forward

Before introducing credit adjustments, we must first establish the baseline, unadjusted price of the derivative. An equity forward is simply a contract that obligates the buyer to purchase a specific stock at a pre-agreed price (the strike) on a specific future date (the maturity). In a standard no-arbitrage framework, the risk-free value of a long forward contract at any time t is determined through a static replication strategy:

$$V_t = S_t - Ke^{-r(T-t)} \quad (51)$$

To recreate the forward's final payoff, an investor can buy one share of the stock today and borrow the present value of the strike price at the risk-free rate r . Because this portfolio and the forward contract give the exact same payoff at maturity, the law of one price states they must have the same value today. Otherwise, there would be an arbitrage opportunity. At inception ($t = 0$), the clean, theoretical fair price is therefore:

$$V_0 = S_0 - Ke^{-rT} \quad (52)$$

To compute this baseline value, we apply our initial market parameters: the last observed stock price from our dataset ($S_0 = 139.0921$), the contractual strike ($K = 150$), the 5-year maturity ($T = 5$), and a constant risk-free rate ($r = 2\%$).

First, discounting the strike yields a present value of:

$$150e^{-0.02 \times 5} = 135.7256 \quad (53)$$

Because the current stock price ($S_0 = 139.0921$) exceeds this discounted strike, applying the pricing formula results in a strictly positive initial contract value:

$$V_0 = 139.0921 - 135.7256 = 3.3665 \quad (54)$$

Economically, this means the Bank's long position has a positive replacement cost today. While this initial value does not guarantee a realized profit at maturity, since the terminal payoff depends entirely on the future stock price S_T , it is critical for our risk analysis. It confirms that from the very first day of the contract, the Bank carries a positive exposure and is immediately subjected to the Counterparty's default risk.

2.2. Monte Carlo estimation of the clean price

In addition to the analytical approach, we also compute the clean fair price using a Monte Carlo simulation. To price derivatives under this numerical framework, we must transition from the real-world historical probability measure ($\mathbb{P}$) used during our calibration to the risk-neutral pricing measure ($\mathbb{Q}$). Under this measure, investors are assumed to be indifferent to risk, meaning the expected return of the underlying stock is no longer its historical drift, but simply the risk-free rate r . Consequently, the continuous dynamics of the stock are governed by the following Stochastic Differential Equation (SDE):

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}} \quad (55)$$

For our numerical simulation, we discretize this process over discrete time steps dt . By applying Itô's Lemma, the exact simulated stock price for the next step is:

$$S_{t+dt} = S_t \exp \left[\left(r - \frac{1}{2}\sigma^2 \right) dt + \sigma \sqrt{dt} Z \right], \quad Z \sim \mathcal{N}(0, 1) \quad (56)$$

For each simulated path, we project the stock price to maturity and compute the terminal payoff $S_T - K$. The Monte Carlo price is then calculated as the discounted average payoff across all M simulations:

$$V_0^{MC} = e^{-rT} \frac{1}{M} \sum_{m=1}^M (S_T^{(m)} - K) \quad (57)$$

To optimize computational efficiency and reduce the variance of our numerical approximation, we implement the method of antithetic variates. For every standard normal shock Z drawn from the distribution, we simultaneously evaluate its perfectly negatively correlated counterpart, $-Z$. This technique effectively halves the number of independent random draws required and significantly stabilizes the convergence of the expected payoff.

Method	Clean forward price
Analytical formula	3.3665
Monte Carlo	3.3667

Table 3: Comparison of the clean fair price of the forward contract.

As demonstrated in Table 3, the numerical Monte Carlo price perfectly converges with the exact analytical value, ensuring our base simulation mechanics are strictly aligned with standard no-arbitrage pricing before introducing bilateral credit risk.

2.3. Exposure profiles: EPE and ENE

While the clean price determines the fair value of the contract at time 0, it provides an incomplete picture of credit risk. A counterparty default can occur at any point before maturity, and the actual financial loss will depend strictly on the contract's mark-to-market value at that exact future moment of default. Because the underlying stock price evolves stochastically, a contract that has a negligible value today could become deeply in-the-money in the future. Therefore, accurately computing CVA and DVA requires us to project and quantify the potential future exposures across the entire life of the contract. At any future time t_i , the simulated mark-to-market value of the forward is given by:

$$V_{t_i} = S_{t_i} - Ke^{-r(T-t_i)} \quad (58)$$

The Expected Positive Exposure (EPE) quantifies the Bank's average financial risk across all future scenarios. It is defined mathematically under the risk-neutral measure as:

$$EPE_{\text{raw}}(t_i) = \mathbb{E}^{\mathbb{Q}} [\max(V_{t_i}, 0)] \quad (59)$$

Because bilateral credit risk is inherently asymmetric, in the event of a default the Bank only suffers a financial loss if the contract has a positive replacement cost at that time (i.e. the Counterparty owes the Bank). If the contract's mark-to-market value is negative, the Counterparty's default does not inflict a credit loss on the Bank. Conversely, the Expected Negative Exposure (ENE) quantifies the Counterparty's financial risk in the event of the Bank's default. Governed by the exact same principle of asymmetric credit risk, the Counterparty only suffers a loss if the contract is a liability for the Bank (i.e., the Bank owes the Counterparty money)([Gregory, 2015](#)).

For graphical purposes, this is often plotted as the expectation of the minimum, $\mathbb{E}^{\mathbb{Q}} [\min(V_{t_i}, 0)]$, to visually denote a liability from the Bank's perspective. However, for the standard DVA

mathematical formulation later, we require the absolute positive magnitude of this exposure. Therefore, the formal raw ENE is defined as:

$$ENE_{\text{raw}}(t_i) = \mathbb{E}^{\mathbb{Q}} [\max(-V_{t_i}, 0)] \quad (60)$$

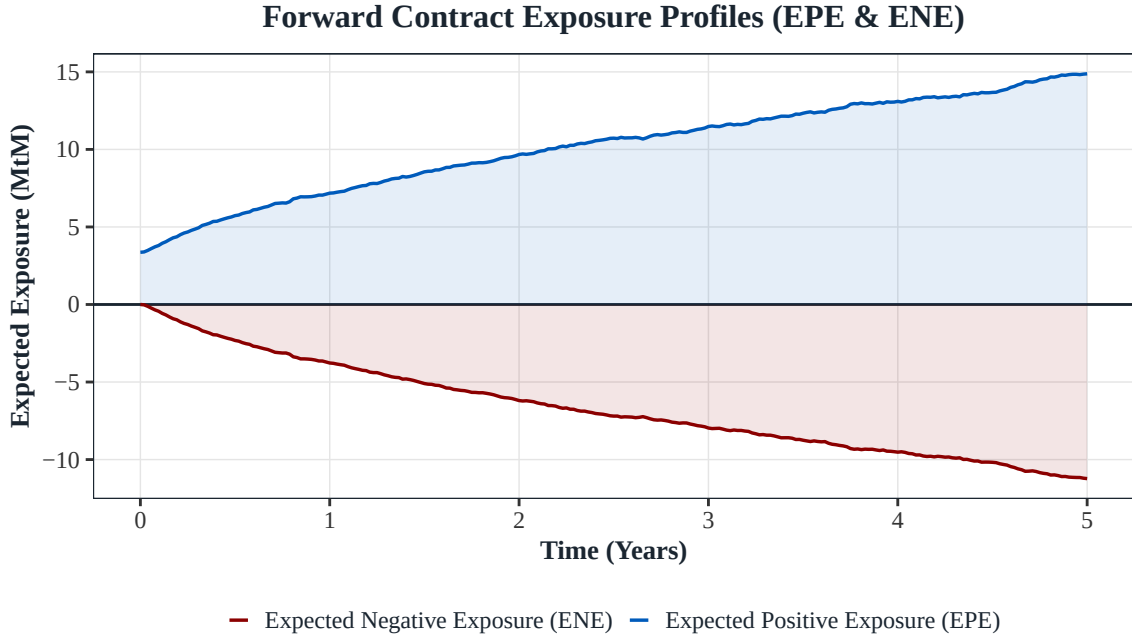


Figure 8: Raw EPE and ENE profiles of the forward contract under independence.

Figure 8 illustrates the evolution of these undiscounted exposure profiles over the 5-year horizon. The blue curve represents the EPE, while the red curve represents the ENE. Both curves appear simultaneously at any given date because the graph is based on thousands of different Monte Carlo scenarios. At a specific future time t_i , some simulated paths will result in a positive forward value, while others will result in a negative value. The model simply separates these outcomes: the blue EPE curve averages only the scenarios where the Bank is owed money, and the red ENE curve averages only the scenarios where the Bank owes money. The smoothness of both curves confirms that our Monte Carlo sample size is sufficiently large to eliminate isolated random noise.

A defining characteristic of these profiles is their continuous expansion over time. This widening is driven by the diffusion of the underlying Geometric Brownian Motion; as time progresses, the accumulated variance of the stock price increases, leading to a much wider distribution of potential mark-to-market swings in both directions.

With the raw exposures defined and stabilized, the final step before calculating the valuation adjustments is to bring these future risks to present value. Because xVA calculations are present-value metrics, the exposures must be discounted back to time zero:

$$EPE(t_i) = e^{-rt_i} \mathbb{E}^{\mathbb{Q}} [\max(V_{t_i}, 0)] \quad (61)$$

and

$$ENE(t_i) = e^{-rt_i} \mathbb{E}^{\mathbb{Q}} [\max(-V_{t_i}, 0)] \quad (62)$$

2.4. CVA and DVA under independence

With the expected exposure profiles (EPE and ENE) now quantified across the duration of the contract, we have the necessary components to adjust the initial risk-free value (V_0) and

determine the true, risk-adjusted fair value ($\tilde{V}_0$). To achieve this, the clean price must be modified by two distinct bilateral credit adjustments: the Credit Valuation Adjustment (CVA) and the Debt Valuation Adjustment (DVA). The relationship is formally defined as:

$$\tilde{V}_0 = V_0 - CVA_0 + DVA_0 \quad (63)$$

The CVA is subtracted from the risk-free price because it represents a projected credit loss for the Bank. Conversely, the DVA is added back because it reflects the theoretical benefit to the Bank, as its own credit deterioration reduces the market value of its future payment obligations. We now compute both adjustments under the baseline assumption that market risk (the underlying stock price) and credit risk (the default probabilities) are statistically independent.

The CVA represents the present value of the expected credit loss to the Bank due to a potential Counterparty default. It can be viewed as an option that pays out the replacement cost of the derivative at the unpredictable moment of default. To calculate this expected loss, we must combine three distinct components at any future time: the likelihood of the default occurring, the size of the Bank's financial exposure, and the fraction of that exposure that cannot be recovered.

A financial loss only happens if the Counterparty defaults when the contract has a positive value (i.e., the Bank is owed money). For this reason, the exposure is floored at zero. Since the exact timing of a bankruptcy is unknown and could occur at any moment dt before maturity T , we must sum these probability-weighted, discounted losses over the entire life of the contract. By integrating these elements, the continuous CVA is defined as ([Gregory, 2015](#)):

$$CVA_0 = LGD_C \int_0^T \mathbb{E}^{\mathbb{Q}} [\max(V_t, 0) e^{-rt}] g_C(t) dt \quad (64)$$

where $LGD_C = 1 - R_C$ is the Loss Given Default and $g_C(t)$ is the Counterparty's default density. In practice, we estimate this continuous integral using a weekly time grid:

$$CVA_0 \approx LGD_C \sum_{i=1}^n EPE(t_i) \Delta PD_C(t_i) \quad (65)$$

Here, $\Delta PD_C(t_i) = G_C(0, t_{i-1}) - G_C(0, t_i)$ represents the probability that the Counterparty defaults within the specific interval $(t_{i-1}, t_i]$, derived directly from its survival probability G_C .

Similarly, the DVA adjusts for the Bank's own default risk. It uses the Expected Negative Exposure (ENE) because the Bank only benefits from its own default when the contract is a liability (i.e., the Bank owes money). The discrete DVA is calculated as:

$$DVA_0 \approx LGD_B \sum_{i=1}^n ENE(t_i) \Delta PD_B(t_i) \quad (66)$$

where $\Delta PD_B(t_i) = G_B(0, t_{i-1}) - G_B(0, t_i)$ is the Bank's marginal probability of default over the time step.

2.5. Analytical computation of CVA and DVA

We compute the valuation adjustments using two methods: an exact analytical approach and a Monte Carlo simulation. We can calculate this analytical baseline because the credit exposure of a forward contract behaves exactly like a standard European option ([Hull, 2018](#)).

While a forward contract can have a negative value, the Bank's credit exposure cannot drop below zero. If the forward's value is positive, the Bank is exposed; if it is negative, the exposure is simply zero. At any future time t , this positive exposure is written as:

$$\max(V_t, 0) = \max(S_t - Ke^{-r(T-t)}, 0) \quad (67)$$

This equation exactly matches the payoff of a European call option expiring at time t , with a strike price of $K_t = Ke^{-r(T-t)}$. Because the mathematical structures are identical, we can compute the expected positive exposure (EPE) directly using the standard Black-Scholes formula:

$$EPE(t) = Call_0(t, K_t) \quad (68)$$

Similarly, the negative exposure (ENE) represents the scenarios where the forward value is negative. Its exposure equation is identical to a European put option:

$$ENE(t) = Put_0(t, K_t) \quad (69)$$

This allows us to calculate the CVA and DVA analytically:

$$CVA_0^{ana} = LGD_C \sum_{i=1}^n Call_0(t_i, K_{t_i}) \Delta PD_C(t_i) \quad (70)$$

and

$$DVA_0^{ana} = LGD_B \sum_{i=1}^n Put_0(t_i, K_{t_i}) \Delta PD_B(t_i) \quad (71)$$

While the analytical method uses the Black-Scholes formula, the Monte Carlo method computes these exact same expectations by averaging the simulated positive and negative exposures across thousands of different stock price paths.

Method	V_0	CVA	DVA	$\tilde{V}_0$
Analytical	3.3665	0.3929	0.0785	3.0521
Monte Carlo	3.3665	0.3976	0.0794	3.0483

Table 4: CVA, DVA, and adjusted value under independence.

Table 4 shows the results. The Monte Carlo simulation gives values that closely match the analytical baseline. The results also show that the CVA is larger than the DVA. This makes sense economically because the Counterparty has higher CDS spreads than the Bank, which means the Counterparty has a higher probability of default. Because the Counterparty is riskier, the expected loss for the Bank (CVA) is larger than the benefit gained from its own default risk (DVA).

Finally, because the CVA is larger than the DVA, the adjusted price ($\tilde{V}_0$) is lower than the clean price (V_0). In other words, the Bank loses more value from the Counterparty's default risk than it gains from its own.

2.6. First-to-default effect

The previous formulas calculated CVA and DVA as if they were entirely separate events. However, in a bilateral OTC contract, only the first default matters. If the Bank defaults first, the contract is immediately terminated and settled. The counterparty default later is no longer relevant and cannot affect the same transaction. Similarly, if the Counterparty defaults first, the Bank's later default does not affect the contract.

To include this first-to-default (FtD) effect, we must look at the joint probability of one party defaulting while the other survives. Because we are still operating under the assumption that the two default processes are statistically independent, this joint probability is simply the product of their individual probabilities. For a Counterparty default to trigger an actual credit loss during the interval $(t_{i-1}, t_i]$, the Counterparty must default during that interval, and the Bank must have survived at least until the beginning of that interval. By multiplying the probabilities of these two independent events, the adjusted metric becomes:

$$\underbrace{\Delta PD_C^{FtD}(t_i)}_{\text{Adjusted probability}} = \underbrace{\Delta PD_C(t_i)}_{\text{Counterparty defaults}} \times \underbrace{G_B(0, t_{i-1})}_{\text{Bank survives}} \quad (72)$$

By the exact same logic, for the Bank's default to trigger a theoretical benefit, the Bank must default during the interval while the Counterparty has survived:

$$\underbrace{\Delta PD_B^{FtD}(t_i)}_{\text{Adjusted probability}} = \underbrace{\Delta PD_B(t_i)}_{\text{Bank defaults}} \times \underbrace{G_C(0, t_{i-1})}_{\text{Counterparty survives}} \quad (73)$$

The first-to-default adjusted CVA and DVA are then computed using these conditional probabilities:

$$CVA_0^{FtD} \approx LGD_C \sum_{i=1}^n EPE(t_i) \Delta PD_C^{FtD}(t_i) \quad (74)$$

and

$$DVA_0^{FtD} \approx LGD_B \sum_{i=1}^n ENE(t_i) \Delta PD_B^{FtD}(t_i) \quad (75)$$

Finally, the bilateral risk-adjusted value of the contract is given by the same formula we had before:

$$\tilde{V}_0^{FtD} = V_0 - CVA_0^{FtD} + DVA_0^{FtD} \quad (76)$$

Method	V_0	CVA (FtD)	DVA (FtD)	$\tilde{V}_0^{FtD}$
Analytical	3.3665	0.3887	0.0754	3.0532
Monte Carlo	3.3665	0.3933	0.0762	3.0494

Table 5: CVA, DVA, and adjusted value with the first-to-default effect.

As shown in Table 5, applying the first-to-default effect reduces both the CVA and the DVA compared to the previous independent calculations. This is because the default of one party only matters if the other party has not already defaulted and thus overlapping default scenarios where both parties eventually default are removed from the calculation. In our specific results, this reduction is relatively small because the baseline default probabilities over a five-year horizon are already quite low, making simultaneous or sequential defaults highly unlikely.

2.7. Conclusion

The clean price of the forward is positive, which means the contract is profitable for the Bank before considering credit risk. However, because there is no central clearing house to guarantee the trade, this profit is at risk. If the Counterparty goes bankrupt when they owe the Bank money, the Bank will not recover the full amount. The CVA is simply the measurement of this expected loss. The DVA works in the exact opposite way. It measures the Bank's own default risk. If the Bank goes bankrupt when the contract has a negative value, it will not fully pay its debt to the Counterparty. In accounting terms, this creates a theoretical benefit that increases the value of the trade for the Bank. In our results, the Counterparty is riskier than the Bank. Because of this, the expected loss (CVA) is larger than the theoretical benefit (DVA). Therefore, the final risk-adjusted value ($\tilde{V}_0$) is strictly lower than the clean price (V_0).

In this section, we have built a complete pricing framework. We calculated the risk-free price, estimated the expected exposures, and applied the baseline credit adjustments. However, this entire baseline model relies on one major assumption: that market risk (the stock price) and credit risk (the default probabilities) are completely independent. While this is a perfect mathematical starting point, it ignores a dangerous real-world problem. What happens if the Counterparty becomes highly likely to default at the exact same time the Bank's exposure is at its highest? To answer this question and measure the true cost of this threat, we will now drop the independence assumption and introduce Wrong-Way Risk (WWR) in the next section.

3. Wrong-way risk, collateral and sensitivity analysis

In this section, we extend the previous analysis by introducing dependence between market risk and credit risk. In Part 2, the stock price, the Bank's default intensity, and the Counterparty's default intensity were assumed to be independent. While this assumption is useful as a benchmark, it is often too simple in practice. When risk factors are correlated, the Bank may face a larger potential loss exactly when the Counterparty is most likely to default. This dynamic, known as Wrong-Way Risk (WWR), increases the CVA and further reduces the value of the trade for the Bank.

We also study the risk-mitigating effect of collateral. We introduce a unilateral Credit Support Annex (CSA) requiring the Counterparty to post collateral every two weeks, provided the required transfer is larger than the Minimum Transfer Amount ($MTA = 5$). We further analyze how changing this margin frequency and the MTA threshold impacts the residual credit exposure. Finally, we perform a comprehensive sensitivity analysis to understand which market and structural inputs have the largest effect on the final valuation adjustments.

3.1. Wrong-way risk and correlated simulation

To include dependence between market risk and credit risk, we jointly simulate the stock price, the bank default intensity and the counterparty default intensity. The objective is to move from the independent framework of the previous section to a model where the three risk factors can move together. We reuse the exact same risk-neutral discrete simulation scheme for the stock price established in Section 2.2:

$$S_{t+dt} = S_t \exp \left[\left(r - \frac{1}{2} \sigma_S^2 \right) dt + \sigma_S \sqrt{dt} Z_t^S \right]. \quad (77)$$

The default intensities are modeled as shifted stochastic processes¹: modeled as shifted stochastic processes¹:

$$\lambda_t^B = \phi^B(t) + y_t^B, \quad \lambda_t^C = \phi^C(t) + y_t^C. \quad (78)$$

The stochastic parts y^B and y^C are simulated using the standard Euler discretization of the Vasicek processes, with the parameters calibrated in Section 1:

$$y_{t+dt}^B = y_t^B + \kappa_B(\theta_B - y_t^B)dt + \eta_B\sqrt{dt}Z_t^B, \quad (79)$$

$$y_{t+dt}^C = y_t^C + \kappa_C(\theta_C - y_t^C)dt + \eta_C\sqrt{dt}Z_t^C. \quad (80)$$

Here, Z_t^S , Z_t^B , and Z_t^C are the standard normal shocks driving the stock price and the respective credit risks. The deterministic shift functions $\phi^B(t)$ and $\phi^C(t)$ are then added to make the model consistent with the survival probability curves bootstrapped from market CDS spreads.

In Section 1, we used the exact analytical AR(1) solution to calibrate the historical data. However, in this section, we switch to the standard Euler discretization for the Monte Carlo simulations. The exact solution was necessary for historical calibration to avoid estimation errors when finding the parameters. For our future simulations, the Euler method is much simpler. It makes it very easy to directly inject our correlated shocks into the equations and because our time step is very small ($dt = 1/52$), the approximation error introduced by the Euler method is negligible.

Because we simulate three risk factors at the same time, we first generate three distinct standard normal shocks at each time step: one for the stock price (Z_t^S), one for the bank default intensity (Z_t^B), and one for the counterparty default intensity (Z_t^C). We group them into a vector of independent shocks:

$$Z_t = \begin{pmatrix} Z_t^S \\ Z_t^B \\ Z_t^C \end{pmatrix}. \quad (81)$$

If we use this vector of independent shocks directly in our equations, our results will be exactly the same as in Part 2. The most important improvement in this section is that we force these shocks to be correlated to properly capture Wrong-Way Risk.

To achieve this, we must apply a mathematical transformation that mixes our independent variables so they match our target correlation matrix ρ . Statistically, this requires finding a transformation matrix L such that $LL^\top = \rho$. While there are multiple mathematical ways to solve this equation, the Cholesky decomposition (Glasserman, 2003) is the most optimal and widely used method for financial Monte Carlo simulations.

$$\rho = \begin{pmatrix} 1 & \rho_{S,B} & \rho_{S,C} \\ \rho_{S,B} & 1 & \rho_{B,C} \\ \rho_{S,C} & \rho_{B,C} & 1 \end{pmatrix} = \begin{pmatrix} 1 & -0.0108 & 0.5155 \\ -0.0108 & 1 & 0.2257 \\ 0.5155 & 0.2257 & 1 \end{pmatrix}. \quad (82)$$

Because our historical correlation matrix is strictly symmetric and positive-definite, the Cholesky algorithm can efficiently factor it into a unique lower-triangular matrix L and its transpose $L^\top$.

¹Please note a small theoretical limitation in our simulation. The exact mathematical formulas assume the default intensity can drop below zero to perfectly match market curves. However, our Monte Carlo code forces the intensities to stay at or above zero ($\lambda_t = \max(\phi(t) + y_t, 0)$) because negative credit risk does not make economic sense. By cutting off the negative values, the model is no longer a perfect Hull-White calibration. Still, this has almost no impact on our final prices because the chance of the intensity actually hitting zero is very low with our specific parameters.

This specific method is highly preferred in quantitative finance for two main reasons. First, it is computationally very fast and numerically stable, which is crucial when running a Monte Carlo engine over thousands of paths. Second, the lower-triangular shape of L applies the correlations sequentially: the first risk factor keeps its independent shock, the second factor combines its shock with the first, and the third factor mixes its shock with the previous two.

We therefore use this lower-triangular matrix L as our exact transformation tool. By multiplying it with our independent vector, we define the final correlated shock vector:

$$\tilde{Z}_t = LZ_t. \quad (83)$$

Once the stock and default intensity paths are fully simulated, we use the intensity paths to generate specific default times. The intensity λ_t^x measures the instantaneous default risk of entity $x \in \{B, C\}$.

For each simulated path, we compute the accumulated default risk up to time t , denoted by Λ_t^x . This quantity strictly increases over time and can be interpreted as the total default pressure that has built up since time 0. Because our simulation moves in discrete time steps, we approximate this continuous integral using a cumulative sum over the time grid:

$$\Lambda_t^x = \int_0^t \lambda_s^x ds \quad \Rightarrow \quad \Lambda_{t_i}^x \approx \sum_{j=1}^i \lambda_{t_{j-1}}^x \Delta t. \quad (84)$$

Next, we must determine the exact moment a default occurs. To do this, we must translate our accumulated default pressure (Λ_t^x) into a probability. In a standard Monte Carlo framework, an event is simulated by drawing a random uniform variable $U \in (0, 1)$, which acts as a randomized probability threshold. Mathematically, the survival probability of an entity at time t is defined as $\exp(-\Lambda_t^x)$. Therefore, a default is triggered the moment this survival probability drops below our random draw:

$$\exp(-\Lambda_t^x) < U \quad (85)$$

By taking the negative natural logarithm of both sides, this condition simplifies to:

$$\Lambda_t^x > -\ln(U) \quad (86)$$

In probability theory, the term $-\ln(U)$ perfectly defines a standard exponential random variable. Following the standard Cox process simulation framework ([Brigo and Mercurio, 2006](#)), we denote this randomized threshold as $\xi_x \sim \text{Exp}(1)$.

By drawing this exponential threshold ξ_x for each entity on each path, we establish a strict breaking point. The exact default time τ_x is defined as the very first moment when the accumulated default pressure overflows this random limit:

$$\tau_x = \inf \{t : \Lambda_t^x > \xi_x\}. \quad (87)$$

This procedure yields one simulated default time for the Bank and one simulated default time for the Counterparty on every single path. These default times are then compared to determine who defaults first. This first-to-default information is the final component required to compute the CVA and DVA path by path.

3.2. CVA and DVA with first-to-default

Now that the joint trajectories of the stock price and the default times are fully simulated, we can compute the CVA and DVA path by path. Because the market exposure and the credit events are generated within the exact same correlated scenario, their interconnected dynamics are preserved.

In this framework, we also include the first-to-default condition. A value adjustment is only recorded if a default occurs before the contract maturity (T). If neither party defaults during the life of the contract, both the CVA and DVA contributions on that specific path are simply zero. If a default does occur, we only consider the entity that defaults first.

For one simulated path m , if the counterparty defaults before the bank and before the maturity of the contract, the bank faces a loss on the positive value of the forward. The CVA for this path is therefore:

$$CVA^{(m)} = LGD_C e^{-r\tau_C^{(m)}} \max \left(V_{\tau_C^{(m)}}^{(m)}, 0 \right), \quad (88)$$

provided that:

$$\tau_C^{(m)} < \tau_B^{(m)} \quad \text{and} \quad \tau_C^{(m)} \leq T. \quad (89)$$

The term $\max \left(V_{\tau_C^{(m)}}^{(m)}, 0 \right)$ means that the bank only suffers a loss when the forward has a positive value at the exact time of the counterparty's default. If the value is negative, the counterparty's default does not create a loss for the bank.

Similarly, if the bank defaults before the counterparty and before maturity, the DVA is computed from the negative value of the contract. On one simulated path, this gives:

$$DVA^{(m)} = LGD_B e^{-r\tau_B^{(m)}} \max \left(-V_{\tau_B^{(m)}}^{(m)}, 0 \right), \quad (90)$$

provided that:

$$\tau_B^{(m)} < \tau_C^{(m)} \quad \text{and} \quad \tau_B^{(m)} \leq T. \quad (91)$$

Here, the term $\max \left(-V_{\tau_B^{(m)}}^{(m)}, 0 \right)$ is used because DVA only matters when the contract has a negative value for the bank. In that case, the bank owes money to the counterparty, and the bank's own default reduces the amount that it will actually pay in full.

The Monte Carlo estimators are obtained by averaging these pathwise quantities over all M simulated scenarios:

$$CVA_0^{WWR} = \frac{1}{M} \sum_{m=1}^M CVA^{(m)}, \quad (92)$$

and

$$DVA_0^{WWR} = \frac{1}{M} \sum_{m=1}^M DVA^{(m)}. \quad (93)$$

The final adjusted value of the forward is then:

$$\tilde{V}_0^{WWR} = V_0 - CVA_0^{WWR} + DVA_0^{WWR}. \quad (94)$$

Case	CVA	DVA	Adjusted price
Independence with first-to-default	0.3887	0.0754	3.0531
Correlation / WWR	0.4495	0.0875	3.0045

Table 6: Impact of correlation and wrong-way risk on CVA, DVA and adjusted price.

Table 6 compares the independent case with the correlated simulation. When correlation is included, the CVA increases from 0.3887 to 0.4495. This means that the counterparty default risk becomes more costly for the bank. Economically, this is a clear sign of Wrong-Way Risk: the bank tends to have a larger positive exposure in the exact scenarios where the counterparty is more likely to default.

The DVA also increases, going from 0.0754 to 0.0875. This means that the bank's own default risk has a slightly larger value in the correlated simulation. However, the increase in CVA is much larger than the increase in DVA. As a result, the adjusted price decreases from 3.0531 to 3.0045.

This result shows that including the dependence between market risk and credit risk reduces the overall value of the trade for the bank. The main reason is that the correlation significantly increases the expected loss due to the counterparty's default.

3.3. Collateral and impact of the CSA parameters

We now introduce collateral according to the Credit Support Annex (CSA). The purpose of collateral is to reduce the financial loss of the bank if the counterparty defaults. If the forward contract has a positive value for the bank, the counterparty must post collateral to cover this exposure.

In our implementation, we assume a unilateral CSA. This means that only the counterparty posts collateral when the contract has a positive value for the bank. They will need to post collateral every two weeks, but a margin call is only made if the required change in collateral is larger than a Minimum Transfer Amount (MTA) of 5.

Without collateral, the exposure used in the CVA computation is simply $\max(V_t, 0)$, which is the positive replacement value of the contract. If the counterparty defaults when this value is positive, the bank may lose a portion of it. With a CSA in place, this exposure is directly reduced by the collateral amount C_t posted by the counterparty:

$$\max(V_t - C_t, 0). \quad (95)$$

Therefore, collateral does not change the fundamental value of the forward itself, but it reduces the amount that remains exposed to counterparty default.

In our simulation, the collateral account is updated only at specific margin dates. At each margin date t , we compute the target collateral, which is simply the positive mark-to-market value of the contract:

$$C_t^{target} = \max(V_t, 0). \quad (96)$$

However, the collateral account is updated only if the required transfer amount strictly exceeds the MTA. More precisely, we compare the new target collateral with the collateral already posted at the previous step (C_{t-}):

$$|C_t^{target} - C_{t-}| > MTA. \quad (97)$$

If this condition is satisfied, a margin call is made and the collateral account is updated to C_t^{target} . If the condition is not satisfied, no transfer is made, and the previous collateral amount is kept. This mechanism creates a residual exposure for the bank because the collateral does not perfectly track the mark-to-market value at every single date.

Case	CVA	DVA	Adjusted price
Without collateral, with WWR	0.4495	0.0875	3.0045
With collateral, $MTA = 5$	0.0269	0.0875	3.4270

Table 7: Impact of collateral on CVA, DVA and adjusted price.

Table 7 shows that collateral strongly reduces the CVA. Without collateral, the CVA is 0.4495. With the two-week collateral rule and $MTA = 5$, the CVA drops to 0.0269. This is a large reduction because most of the bank's positive exposure is already covered by collateral before a possible counterparty default can happen.

The DVA remains unchanged at 0.0875. This is a direct result of the unilateral CSA assumption, since the bank never posts collateral. Because the CVA drops so much, the adjusted price naturally increases from 3.0045 to 3.4270. This is intuitive: the trade becomes much safer for the bank, so the penalty for counterparty credit risk is much smaller.

We also test the impact of the collateral payment frequency (margin dates) and the minimum transfer amount (MTA). Because collateral is not posted continuously in practice, it is important to understand how changing these two rules affects the residual risk for the bank.

Collateral rule	CVA	DVA	Adjusted price
Weekly collateral, $MTA = 5$	0.0229	0.0875	3.4310
Two-week collateral, $MTA = 5$	0.0269	0.0875	3.4270
Monthly collateral, $MTA = 5$	0.0316	0.0875	3.4223
Two-week collateral, $MTA = 0$	0.0099	0.0875	3.4441
Two-week collateral, $MTA = 5$	0.0269	0.0875	3.4270
Two-week collateral, $MTA = 10$	0.0533	0.0875	3.4007

Table 8: Impact of collateral frequency and MTA on CVA, DVA and adjusted price.

Table 8 demonstrates these effects clearly. The first three rows show the impact of the payment frequency while keeping $MTA = 5$. As the time between margin calls increases, from weekly, to two weeks, to monthly, the CVA steadily rises from 0.0229 to 0.0316. This happens because a lower frequency leaves more time for the bank's exposure to grow between two margin dates, making the collateral account less accurate over time.

The last three rows isolate the effect of the MTA, keeping the frequency fixed at two weeks. When $MTA = 0$, every single positive change in required collateral is transferred, resulting in a very low CVA of 0.0099. However, when the MTA increases to 10, the CVA jumps to 0.0533. This confirms that when small margin calls are ignored, a larger portion of the exposure remains uncollateralized, which naturally increases the CVA and decreases the adjusted price.

Overall, collateral has a strong positive effect for the bank because it reduces the expected loss caused by a counterparty default. However, the CVA never drops completely to zero because the collateral is only updated at discrete dates, and the MTA always leaves a small piece of the exposure uncovered.

3.4. Sensitivity analysis

Finally, we perform a sensitivity analysis using a bump-and-revalue approach. We change one parameter at a time and recompute the clean price, CVA, DVA and adjusted price. This allows us to understand which assumptions have the strongest impact on the valuation adjustments. Specifically, we evaluate the sensitivity to the strike price, the maturity, the risk-free rate, the correlation parameters, the estimated parameters of each stochastic process, and the recovery rates. The results in Table 9 illustrate how each individual parameter impacts the final valuation of the trade.

Table 9: Sensitivity analysis of CVA, DVA, and total adjusted price across various market scenarios.

Scenario	Clean Price	CVA	DVA	Adjusted Price	$\Delta\text{CVA_pct}$	$\Delta\text{DVA_pct}$	$\Delta\text{Adj_Price_pct}$
Base case	3.3665	0.4495	0.0875	3.0045	+0.00%	+0.00%	+0.00%
Strike K +5%	-3.4198	0.3137	0.1350	-3.5985	-30.21%	+54.28%	-219.77%
Maturity T +1 year	6.0540	0.6786	0.0941	5.4696	+50.98%	+7.59%	+82.04%
Risk-free rate r +100 bps	9.9859	0.6200	0.0541	9.4199	+37.95%	-38.21%	+213.53%
Correlation $\rho(S, B)$ +0.10	3.3665	0.4489	0.0834	3.0010	-0.14%	-4.68%	-0.12%
Correlation $\rho(S, C)$ +0.10	3.3665	0.4582	0.0875	2.9958	+1.95%	+0.00%	-0.29%
Correlation $\rho(B, C)$ +0.10	3.3665	0.4435	0.0875	3.0105	-1.33%	+0.00%	+0.20%
Stock drift μ_S +10%	3.3665	0.4495	0.0875	3.0045	+0.00%	+0.00%	+0.00%
Stock volatility σ_S +10%	3.3665	0.4867	0.0981	2.9779	+8.28%	+12.11%	-0.89%
Bank κ_B +10%	3.3665	0.4495	0.0881	3.0051	+0.00%	+0.65%	+0.02%
Bank θ_B +10%	3.3665	0.4495	0.0876	3.0046	+0.00%	+0.12%	+0.00%
Bank η_B +10%	3.3665	0.4495	0.0839	3.0010	+0.00%	-4.05%	-0.12%
Counterparty κ_C +10%	3.3665	0.4403	0.0875	3.0137	-2.03%	+0.00%	+0.30%
Counterparty θ_C +10%	3.3665	0.4492	0.0875	3.0048	-0.07%	+0.00%	+0.01%
Counterparty η_C +10%	3.3665	0.4582	0.0875	2.9958	+1.93%	+0.00%	-0.29%
Bank recovery R_B +10 pp	3.3665	0.4495	0.0795	2.9965	+0.00%	-9.12%	-0.27%
Counterparty recovery R_C +10 pp	3.3665	0.4358	0.0872	3.0179	-3.05%	-0.35%	+0.45%

The strike price K has a very strong impact on the contract's value. When K increases by 5%, the clean price switches from positive to negative, going from 3.3665 to -3.4198. Because the contract is now less favorable for the bank, the positive exposure becomes much smaller. As a result, the CVA drops by 30.21% because the bank is less likely to receive money. At the same time, the Expected Negative Exposure (ENE) increases, pushing the DVA up by 54.28%. This happens because a higher strike price makes the contract more favorable to the counterparty, meaning the bank is expected to owe them more money. Together, these changes lead to a large 219.77% drop in the total adjusted price.

Extending the maturity T by 1 year increases both credit adjustments. This is expected because a longer time horizon gives the stock price more time to reach extreme values, and it also increases the total probability that a default happens before the end of the contract. As a result, the CVA goes up by 50.98% and the DVA goes up by 7.59%. The net adjusted price increases by 82.04% simply because the underlying forward value becomes much larger over a longer period.

The risk-free rate r also strongly affects the valuation. An increase of 100 basis points lowers the present value of the strike payment. Since we hold a long forward position, this change

makes the contract significantly more valuable for the bank, pushing the clean price up to 9.9859. Consequently, the CVA increases by 37.95%, driven by two opposing forces. First, a higher r increases the drift in the Monte Carlo simulation, causing the simulated stock prices S_t to grow faster. This leads to a much higher forward value V_t on almost every path, greatly expanding the positive exposure. Second, the higher rate decreases the discount factor e^{-rt} , which would normally reduce the CVA. However, the rapid growth of the stock prices easily overpowers this minor discounting effect, resulting in a large net increase in both the expected positive exposure and the final CVA.

On the other hand, the DVA drops by 38.21%. This happens for the exact opposite reasons. First, the higher rate r lowers the discount factor, which slightly reduces the Expected Negative Exposure (ENE). Second, and more importantly, the higher drift pushes the simulated stock prices up. Because the contract value V_t becomes deeply positive, the formula for negative exposure, $\max(-V_t, 0)$, will simply return 0 in most scenarios. In economic terms, this means the bank expects to owe almost nothing to the counterparty, so the bank's own default risk loses its theoretical value. This simulation effect is very strong and drives the final DVA down.

Finally, the total adjusted price increases by a massive 213.53%. This happens mainly because the higher risk-free rate lowers the present value of the strike, which makes the clean price jump. This positive gain on the basic contract is so large that it easily outweighs the changes in CVA and DVA, which act as relatively small corrections in comparison.

The stock drift parameter μ_S has very low sensitivity on any of the outputs. This makes perfect sense because our simulation engine uses the risk-neutral pricing measure. In a risk-neutral world, the stock is forced to grow at the risk-free rate r , meaning the historical drift parameter μ_S is completely ignored by the math.

In contrast, the stock volatility σ_S has a strong impact on both credit adjustments. When σ_S increases by 10%, the CVA rises by 8.28% and the DVA increases by 12.11%. This happens because higher volatility creates a much wider spread of future stock prices S_t . This wider distribution naturally increases our exposure in both directions. On the upside, higher simulated stock prices create larger positive values for V_t , which increases the expected positive exposure and the final CVA. On the downside, lower stock prices create deeper negative values for V_t , which increases the expected negative exposure and the final DVA. Ultimately, since our starting CVA is much larger than our starting DVA, the extra cost of the CVA easily outweighs the extra benefit of the DVA. This causes the total adjusted price to decrease by 0.89%.

The correlation parameters have a smaller but logical impact on the final valuation. Increasing the stock-counterparty correlation $\rho(S, C)$ by +0.10 raises the CVA by 1.95%. This is a classic example of wrong-way risk: the counterparty becomes more likely to default in the exact scenarios where the contract has a large positive value for the bank. On the other hand, increasing the stock-bank correlation $\rho(S, B)$ by +0.10 lowers the DVA by 4.68%. This creates a right-way risk effect for the bank's own liabilities. In scenarios where the contract value is negative (meaning the bank owes money), the bank's default risk actually goes down, which reduces the theoretical benefit of the DVA. Finally, increasing the correlation between the bank and the counterparty $\rho(B, C)$ by +0.10 causes a small 1.33% decrease in the CVA. This is driven by the first-to-default rule: if the two entities are highly correlated, they are more likely to experience distress simultaneously. If the bank defaults just before or alongside the counterparty, the CVA exposure is cancelled, slightly reducing the expected charge.

The default intensity parameters for the bank ($\kappa_B, \theta_B, \eta_B$) change the DVA but leave the CVA completely untouched. Because we utilize a deterministic shift function $\phi^B(t)$ that perfectly calibrates to market curves, bumping the underlying Vasicek parameters has a very muted economic effect. The shift function automatically absorbs the majority of the parameter shocks

during the re-calibration phase to keep the initial survival probabilities constant. The minor variations we observe in the DVA (such as the 4.05% drop when volatility η_B increases) are simply non-linear simulation effects caused by flooring the intensities at zero during the Monte Carlo paths.

In the same way, the counterparty's parameters (κ_C , θ_C , η_C) only impact the CVA and leave the DVA unchanged. Increasing the counterparty's mean-reversion speed κ_C by 10% causes a 2.03% drop in the CVA because it reduces the variance of the counterparty's credit risk over time. Shifting the long-term average θ_C upward by 10% causes almost no change (-0.07%) in the CVA, as the exact piecewise function $\phi^C(t)$ completely offsets this adjustment to match the market credit curves. Finally, increasing the counterparty's volatility η_C by 10% increases the CVA by 1.93%, as larger shocks in the unfloored credit intensity naturally increase the upside tail risk of default over the 5-year life of the contract.

The recovery rate parameters scale our final credit adjustments by changing the Loss Given Default multiplier ($LGD = 1 - R$). When testing these parameters, we observe a clear difference between the bank (R_B) and the counterparty (R_C). This difference comes from a trade-off between a direct effect (the lower LGD) and an indirect effect (the model re-calibration).

For the counterparty recovery rate (R_C), the direct effect is stronger. An upward shift of R_C by 10 percentage points causes a 3.05% drop in the CVA. Even though the re-bootstrapping mechanism forces the underlying default intensities λ_C to increase to match the market CDS spreads, this extra risk is safely covered by our biweekly collateral margin calls. Because our positive exposure is well-protected by the CSA, the simple benefit of a lower LGD dominates, naturally reducing the CVA.

For the bank recovery rate (R_B), an upward shift of 10 percentage points causes a 9.12% drop in the DVA. Because our contract assumes a unilateral CSA where the bank does not post any collateral, our negative exposure remains completely unprotected. When a higher R_B forces the re-bootstrap algorithm to sharply increase the bank's default intensities λ_B to keep matching the market CDS spreads, this higher default probability fights directly against the mathematical decrease in LGD_B . Because the DVA calculation essentially multiplies these two opposing forces together, they largely cancel each other out. The 9.12% drop is simply the small net difference left over. This proves that recovery rate shocks must always be interpreted together with the model's re-calibration effects.

3.5. Conclusion

Including correlation makes the valuation more realistic because market exposure and default risk move together. In our framework, this dependency increases both the CVA and the DVA. However, because the counterparty carries higher credit risk spreads than the bank, the absolute increase in the CVA is larger. As a result, the total adjusted price decreases compared to the independent benchmark, clearly showing the tangible cost of wrong-way risk. While an independent model assumes that market movements and defaults happen randomly and separately, our correlated framework captures the true economic danger: the bank's positive exposure peaks exactly when the counterparty is most likely to fail. This dangerous overlap forces a heavier CVA penalty, proving that ignoring these dependencies artificially inflates the value of the trade.

Collateral acts as a highly effective risk mitigant. Introducing a biweekly margin call under a unilateral CSA with $MTA = 5$ drops the CVA from 0.4495 to 0.0269. This significant reduction proves that frequent collateralization successfully secures most of the bank's positive exposure. Nevertheless, the CVA does not drop completely to zero because collateral updates only occur at discrete dates, and the MTA threshold always leaves a minor residual exposure uncovered.

Finally, the sensitivity analysis confirms that the primary drivers of the total adjusted price are the strike, maturity, risk-free rate, stock volatility, and recovery rate assumptions. These outputs demonstrate an important concept: xVA is not merely an isolated credit calculation. While structural parameters like recovery rates trigger substantial credit shifts via re-calibration, the final valuation remains deeply driven by the underlying market dynamics of the contract.

AI USAGE DISCLOSURE

During the preparation of this report, the authors used AI tools to assist with code debugging, the refinement of technical scripts, and the optimization of document structure and phrasing. These tools served as an editorial and technical aid. The authors have maintained full oversight throughout the process; we affirm our comprehensive understanding of the methodology, the mathematical derivations, and the interpretations presented, and we take complete responsibility for the final content of this work.

References

- Black, F., & Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
- Brigo, D., & Mercurio, F. (2006). *Interest rate models-theory and practice: With smile, inflation and credit*. Springer Science & Business Media.
- Brigo, D., & Vrins, F. (2018). Disentangling wrong-way risk: Pricing CVA via change of measures and drift adjustment. *European Journal of Operational Research*, 269(3), 1154–1164.
- Glasserman, P. (2003). *Monte Carlo methods in financial engineering* (Vol. 53). Springer.
- Gregory, J. (2015). *The xVA challenge: Counterparty credit risk, funding, collateral, and capital* (3rd). John Wiley & Sons.
- Hull, J. (2018). *Options, futures, and other derivatives* (10th ed.). Pearson.
- Hull, J., & White, A. (1990). Pricing interest-rate-derivative securities. *The Review of Financial Studies*, 3(4), 573–592.
- Ibe, O. C. (2013). *Elements of random walk and diffusion processes*. John Wiley & Sons.
- Shreve, S. E. (2004). *Stochastic calculus for finance II: Continuous-time models*. Springer Finance.
- Vasicek, O. (1977). An equilibrium characterization of the term structure. *Journal of Financial Economics*, 5(2), 177–188.
- Vrins, F. (2025). *Derivatives pricing*.